Course: IT3021 – Data Warehousing and Business Intelligence

Assignment 01

Student Number – IT23621060

Batch – Kandy UNI

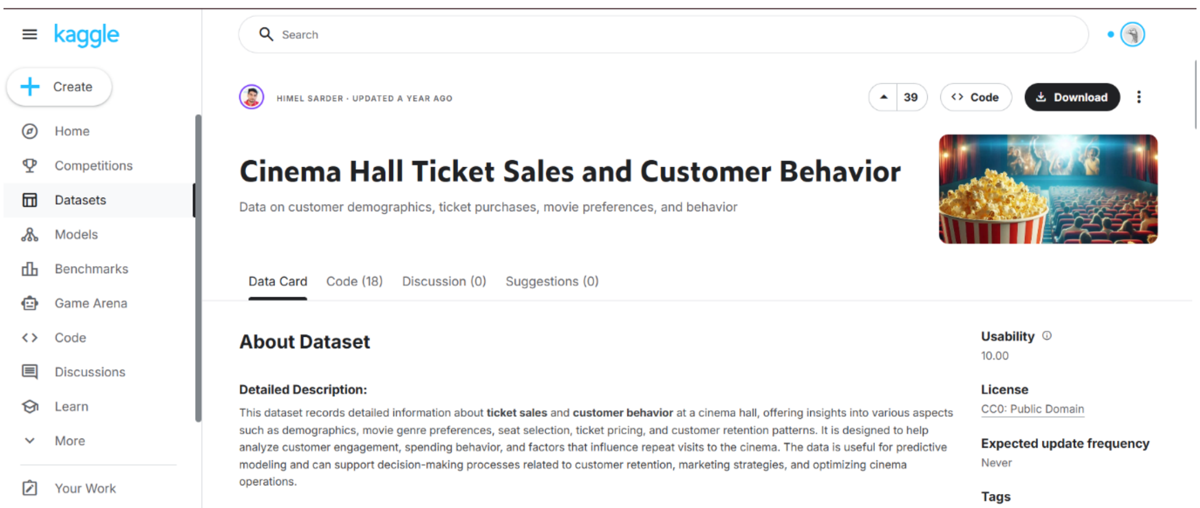
Step 1 – Data set selection

I have selected a dataset that represents a cinema ticket booking system, including customer details, movie information hall and seat allocation, and ticket transactions. The dataset captures booking dates, prices, and quantities, allowing analysis of sales and customer behavior.

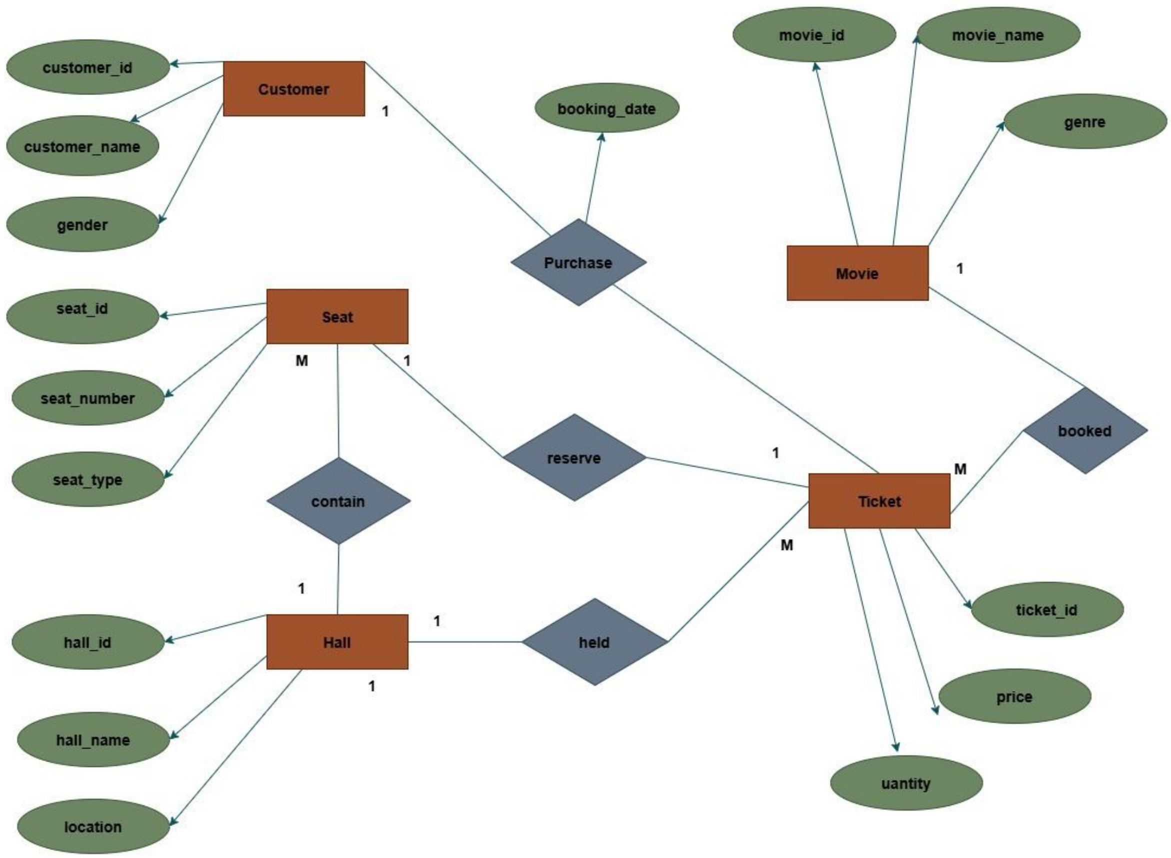
This dataset represents an OLTP system as it contains transactional data with multiple related entities and frequent updates. It is suitable for designing a data warehouse solution.

Data set: - Cinema Hall Ticket Sales and Customer Behavior
https://www.kaggle.com/datasets/himelsarder/cinema-hall-ticket-sales-and-customer-behavior

The dataset was organized into multiple files such as customers, movies, halls, seats, and ticket transactions, Necessary modifications and extensions were applied to create a consistent and suitable dataset for the data warehouse implementation.



ER Diagram:


customer_id
Customer
1
customer_name
gender
booking_date
movie_id
movie_name
genre
Purchase
Movie
1
seat_id
Seat
M
1
seat_number
seat_type
reserve
1
booked
M
contain
Ticket
M
1
1
hall_id
Hall
held
ticket_id
1
hall_name
price
location
uantity

Step 2: Preparation of data sources

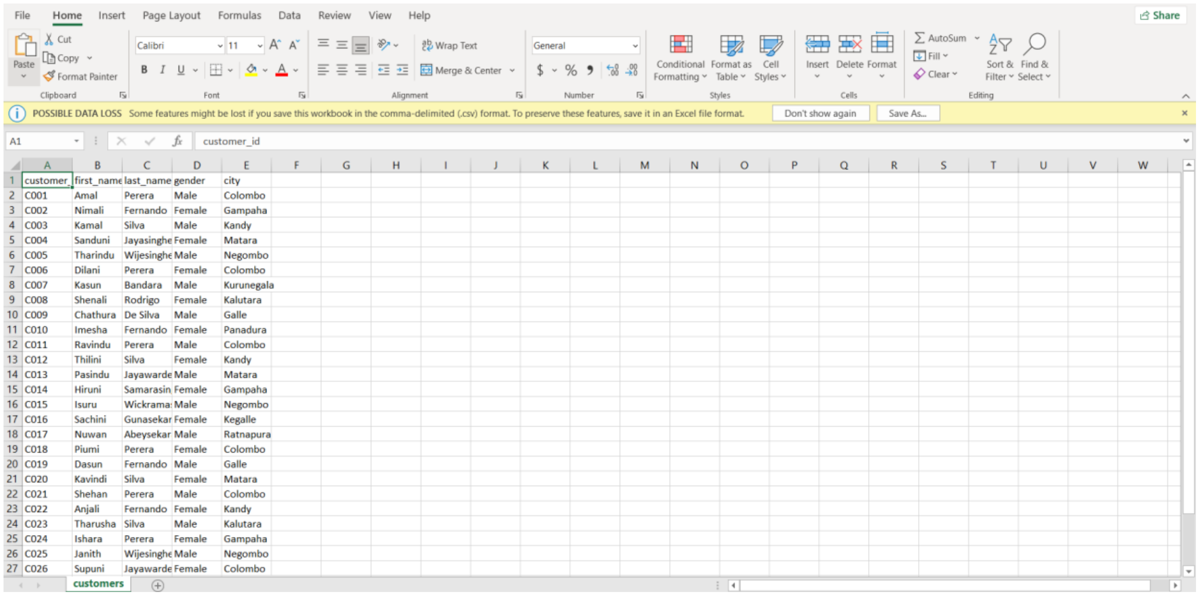
The original dataset was designed to simulate a cinema ticket booking system, containing transactional and descriptive data related to customers, movies, halls, seats and ticket purchase. The dataset initially existed in a semi-structured format with combined attributes, which required transformation to make it suitable for a data warehouse solution.

To prepare the data for ETL processing, the dataset was cleaned, normalized and split into multiple structured files based on entities. The following data sources were created.

- Customers dataset– containing customer details such as customer ID, name, gender and city.
- Movies dataset- containing movie-related information such as movie ID, movie name and genre.
- Halls dataset- containing hall details including hall ID, hall name, and location.
- Seats dataset- containing seat information such as seat ID, seat number, seat type and associated hall.
- Tickets dataset- representing transactional data, including ticket ID, customer ID, movie ID, hall ID, seat ID, booking date, price, and quantity.

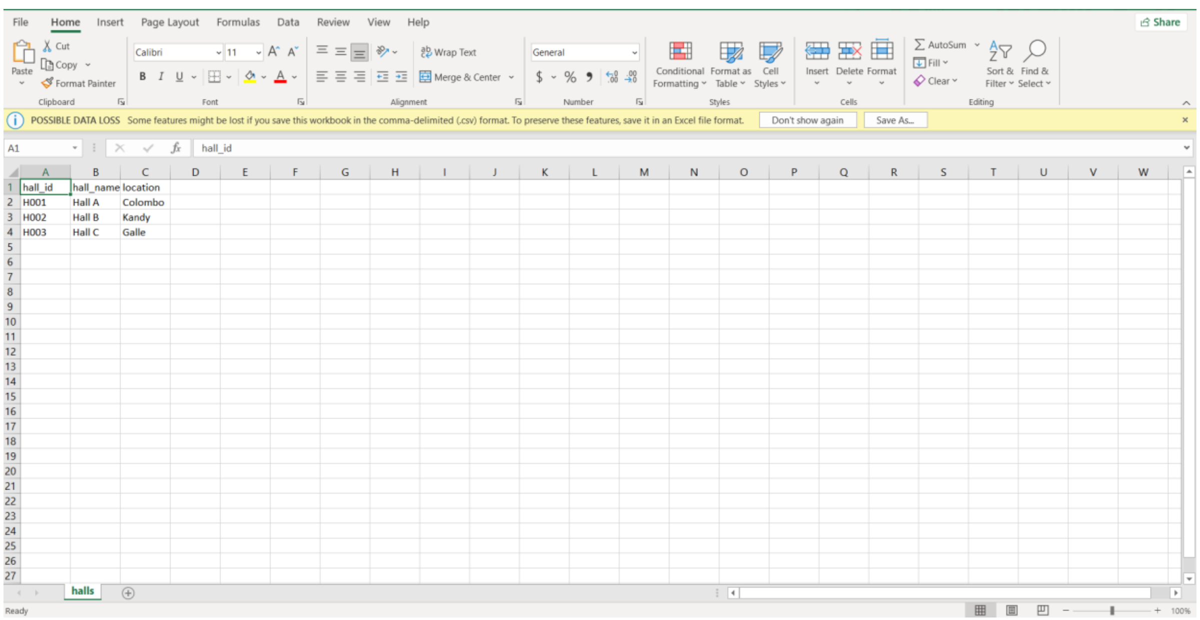
The final dataset structure represents a normalized OLTP-style system, which serves as the input for the ETL process and is later transformed into a star schema for analytical purpose.

During data preparation, transformations such as data cleaning, handling missing values, and standardizing data types were performed. Additionally, derived attributes such as date_key were created to support dimensional modeling.



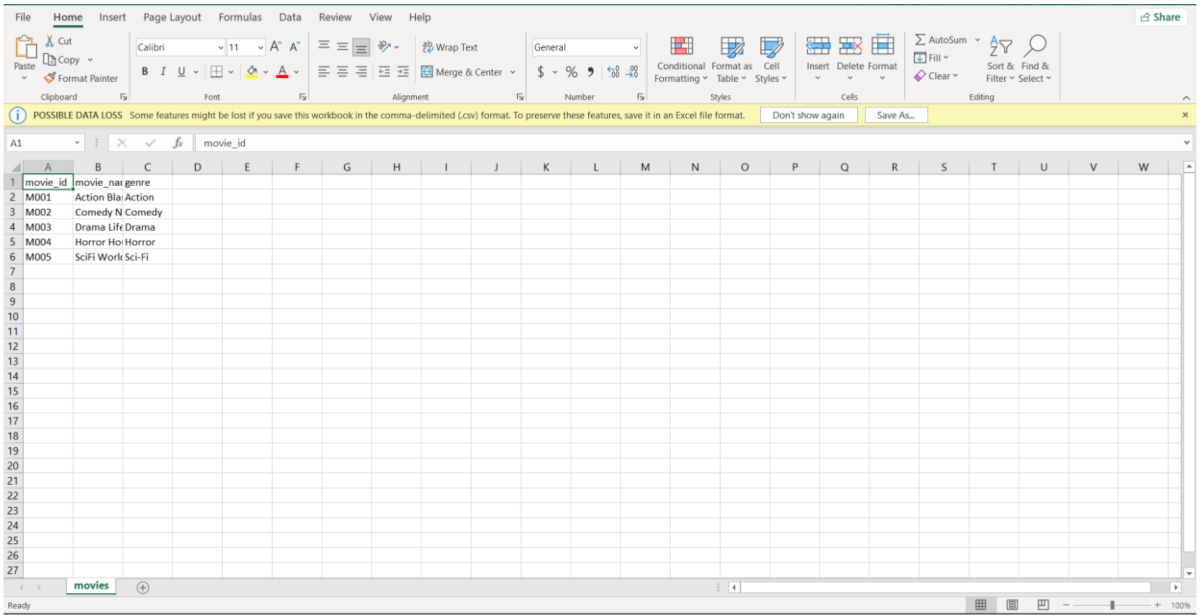
customer_	first_name	last_name	gender	city
C001	Amal	Perera	Male	Colombo
C002	Nimali	Fernando	Female	Gampaha
C003	Kamal	Silva	Male	Kandy
C004	Sanduni	Jayasinghe	Female	Matara
C005	Tharindu	Wijesinghe	Male	Negombo
C006	Dilani	Perera	Female	Colombo
C007	Kasun	Bandara	Male	Kurunegala
C008	Shenali	Rodrigo	Female	Kalutara
C009	Chathura	De Silva	Male	Galle
C010	Imesha	Fernando	Female	Panadura
C011	Ravindu	Perera	Male	Colombo
C012	Thilini	Silva	Female	Kandy
C013	Pasindu	Jayawarde	Male	Matara
C014	Hiruni	Samarasin	Female	Gampaha
C015	Isuru	Wickrama	Male	Negombo
C016	Sachini	Gunasekar	Female	Kegalle
C017	Nuwan	Abeysekar	Male	Ratnapura
C018	Piumi	Perera	Female	Colombo
C019	Dasun	Fernando	Male	Galle
C020	Kavindi	Silva	Female	Matara
C021	Shehan	Perera	Male	Colombo
C022	Anjali	Fernando	Female	Kandy
C023	Tharusha	Silva	Male	Kalutara
C024	Ishara	Perera	Female	Gampaha
C025	Janith	Wijesinghe	Male	Negombo
C026	Supuni	Jayawarde	Female	Colombo

Customers.csv

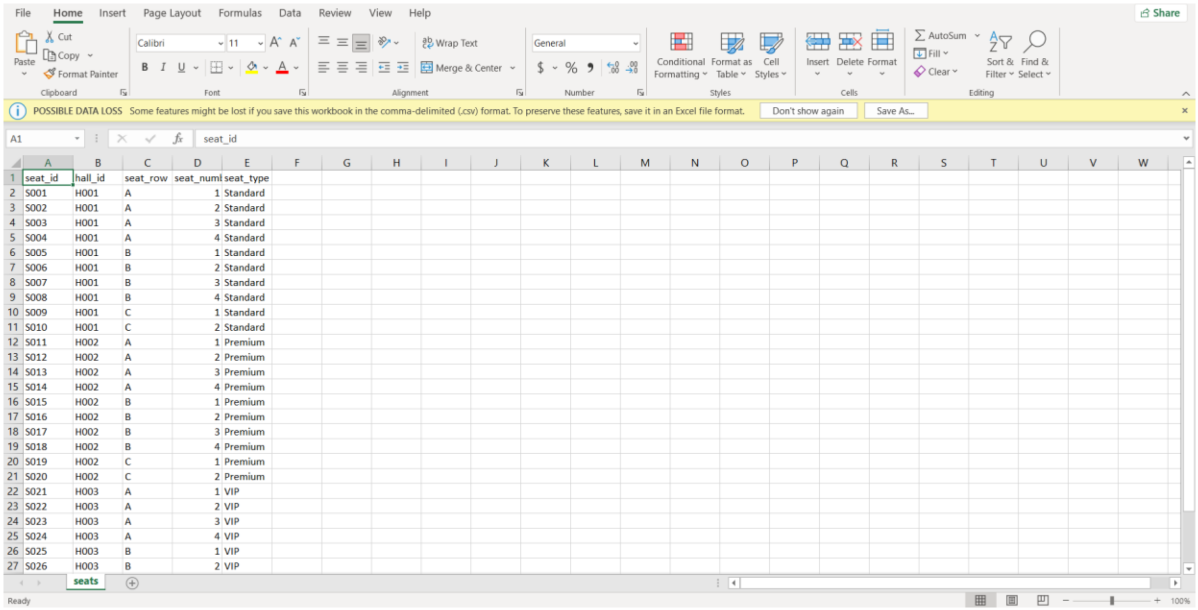


hall_id	hall_name	location
H001	Hall A	Colombo
H002	Hall B	Kandy
H003	Hall C	Galle

Halls.csv

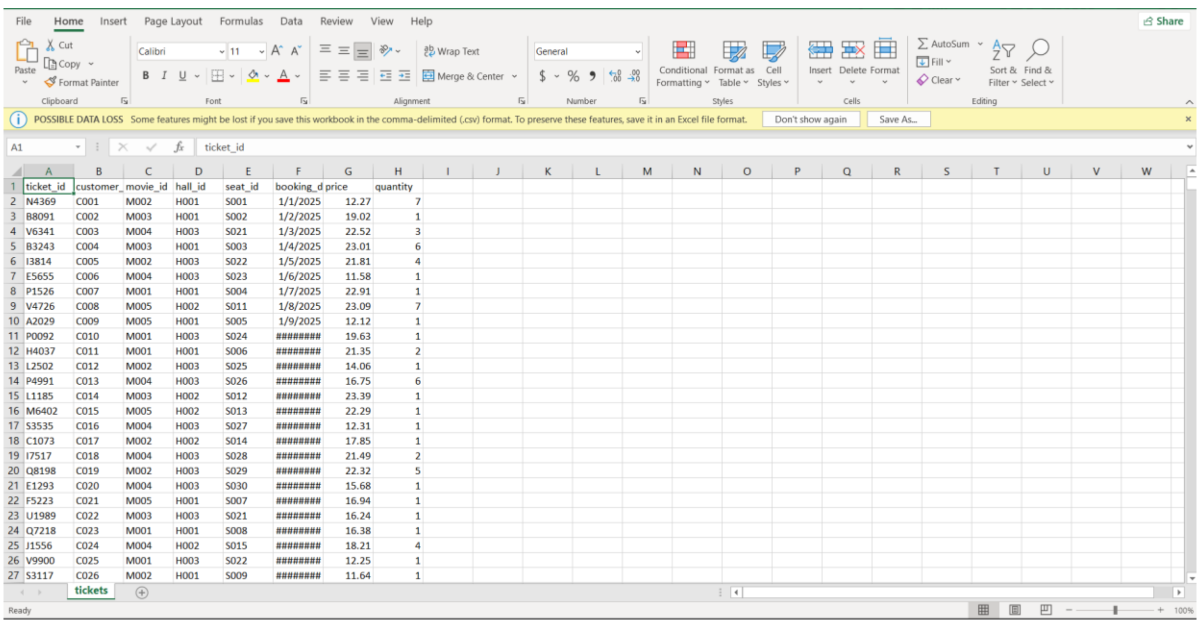

POSSIBLE DATA LOSS Some features might be lost if you save this workbook in the comma-delimited (.csv) format. To preserve these features, save it in an Excel file format.
movie_id
movie_nar
genre
M001
Action Bla
Action
M002
Comedy N
Comedy
M003
Drama Life
Drama
M004
Horror Ho
Horror
M005
SciFi Work
Sci-Fi
movies

Movies.csv



	A	B	C	D	E
1	seat_id	hall_id	seat_row	seat_numl	seat_type
2	S001	H001	A	1	Standard
3	S002	H001	A	2	Standard
4	S003	H001	A	3	Standard
5	S004	H001	A	4	Standard
6	S005	H001	B	1	Standard
7	S006	H001	B	2	Standard
8	S007	H001	B	3	Standard
9	S008	H001	B	4	Standard
10	S009	H001	C	1	Standard
11	S010	H001	C	2	Standard
12	S011	H002	A	1	Premium
13	S012	H002	A	2	Premium
14	S013	H002	A	3	Premium
15	S014	H002	A	4	Premium
16	S015	H002	B	1	Premium
17	S016	H002	B	2	Premium
18	S017	H002	B	3	Premium
19	S018	H002	B	4	Premium
20	S019	H002	C	1	Premium
21	S020	H002	C	2	Premium
22	S021	H003	A	1	VIP
23	S022	H003	A	2	VIP
24	S023	H003	A	3	VIP
25	S024	H003	A	4	VIP
26	S025	H003	B	1	VIP
27	S026	H003	B	2	VIP

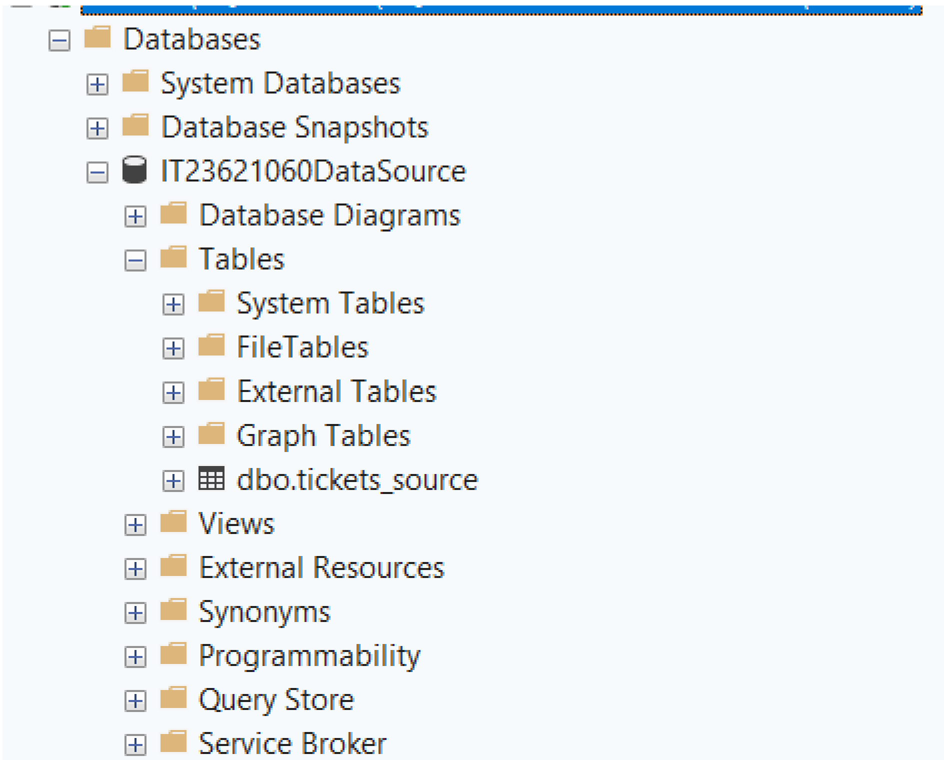
Seats.csv



	A	B	C	D	E	F	G	H
1	ticket_id	customer_	movie_id	hall_id	seat_id	booking_d	price	quantity
2	N4369	C001	M002	H001	S001	1/1/2025	12.27	7
3	B8091	C002	M003	H001	S002	1/2/2025	19.02	1
4	V6341	C003	M004	H003	S021	1/3/2025	22.52	3
5	B3243	C004	M003	H001	S003	1/4/2025	23.01	6
6	I3814	C005	M002	H003	S022	1/5/2025	21.81	4
7	E5655	C006	M004	H003	S023	1/6/2025	11.58	1
8	P1526	C007	M001	H001	S004	1/7/2025	22.91	1
9	V4726	C008	M005	H002	S011	1/8/2025	23.09	7
10	A2029	C009	M005	H001	S005	1/9/2025	12.12	1
11	P0092	C010	M001	H003	S024	########	19.63	1
12	H4037	C011	M001	H001	S006	########	21.35	2
13	L2502	C012	M002	H003	S025	########	14.06	1
14	P4991	C013	M004	H003	S026	########	16.75	6
15	L1185	C014	M003	H002	S012	########	23.39	1
16	M6402	C015	M005	H002	S013	########	22.29	1
17	S3535	C016	M004	H003	S027	########	12.31	1
18	C1073	C017	M002	H002	S014	########	17.85	1
19	I7517	C018	M004	H003	S028	########	21.49	2
20	Q8198	C019	M002	H003	S029	########	22.32	5
21	E1293	C020	M004	H003	S030	########	15.68	1
22	F5223	C021	M005	H001	S007	########	16.94	1
23	U1989	C022	M003	H003	S021	########	16.24	1
24	Q7218	C023	M001	H001	S008	########	16.38	1
25	J1556	C024	M004	H002	S015	########	18.21	4
26	V9900	C025	M001	H003	S022	########	12.25	1
27	S3117	C026	M002	H001	S009	########	11.64	1

Tickets.csv

Source Transaction Table in SQL Server:



The prepared data sources were used as inputs for the SSIS-based ETL process to load data into the data warehouse.

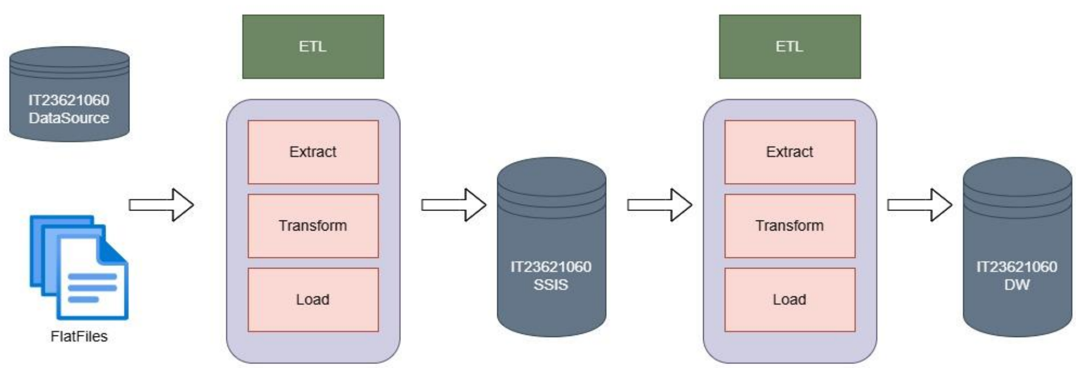
Step 3: Solution architecture

The overall architecture of the solution is designed to integrate multiple data sources and transform them into a structured data warehouse for analytical processing.

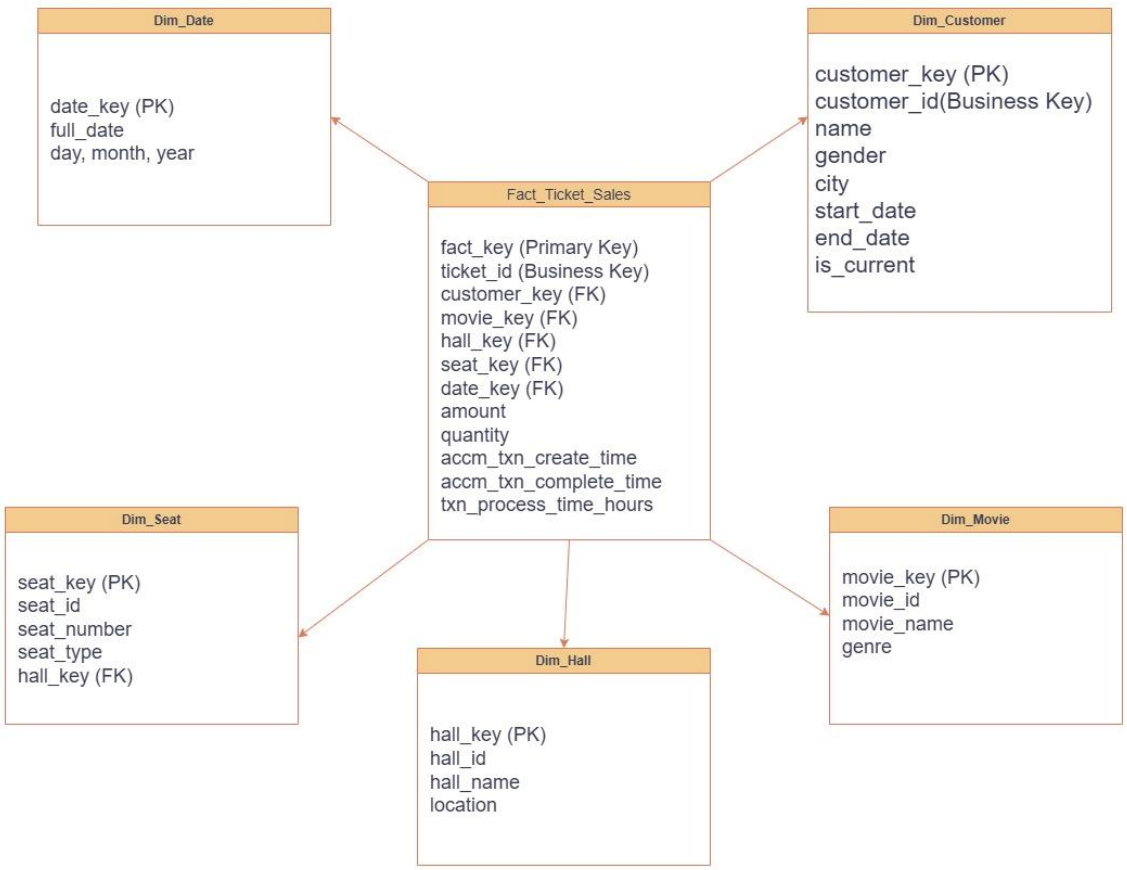
The system consists of three main layers:

- Data source layer – The data originates from multiple sources. (Flat files, SQL server database)
- ETL layer – The extract, Transform, Load (ETL) process is implemented using SQL server integration services. (SSIS)
- Data warehouse layer – The processed data is stored in a SQL Server Data Warehouse (IT23621060DW) using a star schema.

The ETL layer ensures data consistency and integrates data from multiple sources into a unified data warehouse.



Step 4: Data warehouse design & development



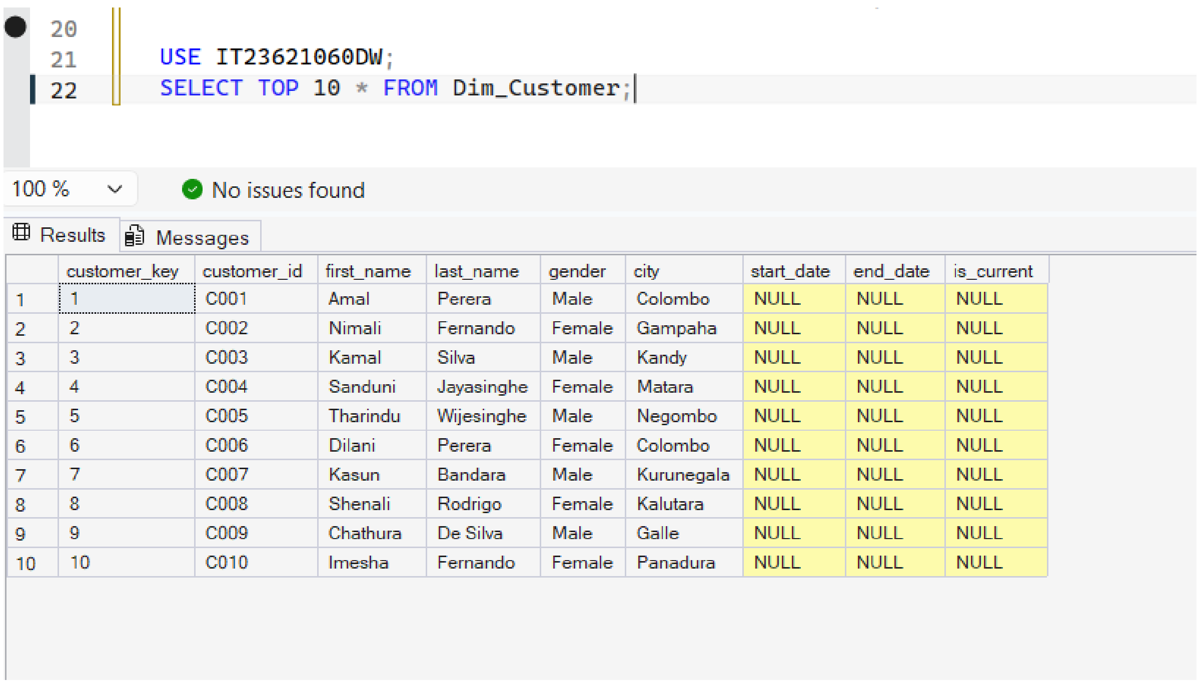
The data warehouse was designed using a star schema, consisting of a central fact table (Fact_Ticket_Sales) connected to multiple dimension tables (Dim_Customer, Dim_Movie, Dim_Hall, Dim_Seat, and Dim_Date).

The star schema structure enables efficient storage and fast retrieval of ticket transaction data, making it ideal for analyzing cinema booking patterns and supporting business intelligence operations.

Fact Table -> The Fact_Ticket_Sales table stores transactional data including ticket details, surrogate keys, amount, quantity and timestamps.

Grain -> Each record in the fact table represents one ticket transaction for a specific customer, movie, hall, seat and booking date.

- Each ticket_id represents a unique transaction.
- A customer can purchase multiple tickets over time.
- Each ticket is associated with one movie, one hall, one seat, and one booking date.
- Seat belongs to a specific hall and does not change.
- Booking date is used to derive a date_key for time-based analysis.
- Transaction completion details (completion time and processing duration) are updated later, supporting an accumulating fact table.



```sql
USE IT23621060DW;
SELECT TOP 10 * FROM Dim_Customer;
```


	customer_key	customer_id	first_name	last_name	gender	city	start_date	end_date	is_current
1	1	C001	Amal	Perera	Male	Colombo	NULL	NULL	NULL
2	2	C002	Nimali	Fernando	Female	Gampaha	NULL	NULL	NULL
3	3	C003	Kamal	Silva	Male	Kandy	NULL	NULL	NULL
4	4	C004	Sanduni	Jayasinghe	Female	Matara	NULL	NULL	NULL
5	5	C005	Tharindu	Wijesinghe	Male	Negombo	NULL	NULL	NULL
6	6	C006	Dilani	Perera	Female	Colombo	NULL	NULL	NULL
7	7	C007	Kasun	Bandara	Male	Kurunegala	NULL	NULL	NULL
8	8	C008	Shenali	Rodrigo	Female	Kalutara	NULL	NULL	NULL
9	9	C009	Chathura	De Silva	Male	Galle	NULL	NULL	NULL
10	10	C010	Imesha	Fernando	Female	Panadura	NULL	NULL	NULL

```sql
CREATE TABLE Dim_Customer (
    customer_key INT IDENTITY(1,1) PRIMARY KEY,
    customer_id VARCHAR(10),
    first_name VARCHAR(50),
    last_name VARCHAR(50),
    gender VARCHAR(10),
    city VARCHAR(50),
    start_date DATETIME,
    end_date DATETIME,
    is_current BIT
);

CREATE TABLE Dim_Movie (
    movie_key INT IDENTITY(1,1) PRIMARY KEY,
    movie_id VARCHAR(10),
    movie_name VARCHAR(100),
    genre VARCHAR(50)
);
```

```sql
CREATE TABLE Dim_Hall (
    hall_key INT IDENTITY(1,1) PRIMARY KEY,
    hall_id VARCHAR(10),
    hall_name VARCHAR(50),
    location VARCHAR(50)
);

CREATE TABLE Dim_Seat (
    seat_key INT IDENTITY(1,1) PRIMARY KEY,
    seat_id VARCHAR(10),
    hall_id VARCHAR(10),
    seat_row VARCHAR(10),
    seat_number INT,
    seat_type VARCHAR(20)
);

CREATE TABLE Dim_Date (
    date_key INT PRIMARY KEY,
    full_date DATE,
    day_no INT,
    month_no INT,
    year_no INT
);
```

```sql
CREATE TABLE Fact_Ticket_Sales (
    fact_key INT IDENTITY(1,1) PRIMARY KEY,
    ticket_id VARCHAR(50),
    customer_key INT,
    movie_key INT,
    hall_key INT,
    seat_key INT,
    date_key INT,
    amount DECIMAL(10,2),
    quantity INT,
    accm_txn_create_time DATETIME,
    accm_txn_complete_time DATETIME,
    txn_process_time_hours INT
);
```

Column Name	Data Type	Allow Nulls
customer_key	int	☐
customer_id	varchar(10)	☑
first_name	varchar(50)	☑
last_name	varchar(50)	☑
gender	varchar(10)	☑
city	varchar(50)	☑
start_date	datetime	☑
end_date	datetime	☑
is_current	bit	☑
		☐

Column Name	Data Type	Allow Nulls
movie_key	int	☐
movie_id	varchar(10)	☑
movie_name	varchar(100)	☑
genre	varchar(50)	☑
		☐

Column Name	Data Type	Allow Nulls
hall_key	int	☐
hall_id	varchar(10)	☑
hall_name	varchar(50)	☑
location	varchar(50)	☑
		☐

Column Name	Data Type	Allow Nulls
seat_key	int	☐
seat_id	varchar(10)	☑
hall_id	varchar(10)	☑
seat_row	varchar(10)	☑
seat_number	int	☑
seat_type	varchar(20)	☑
		☐

Column Name	Data Type	Allow Nulls
date_key	int	☐
full_date	date	☑
day_no	int	☑
month_no	int	☑
year_no	int	☑
		☐

Column Name	Data Type	Allow Nulls
fact_key	int	☐
ticket_id	varchar(50)	☑
customer_key	int	☑
movie_key	int	☑
hall_key	int	☑
seat_key	int	☑
date_key	int	☑
amount	decimal(10, 2)	☑
quantity	int	☑
accm_txn_create_time	datetime	☑
accm_txn_complete_time	datetime	☑
txn_process_time_hours	int	☑
		☐

Step 5: ETL development

The ETL (Extract, Transform, Load) process was implemented using SQL Server Integration Services (SSIS) to load data from multiple sources into the data warehouse.

The ETL process was divided into two main stages:

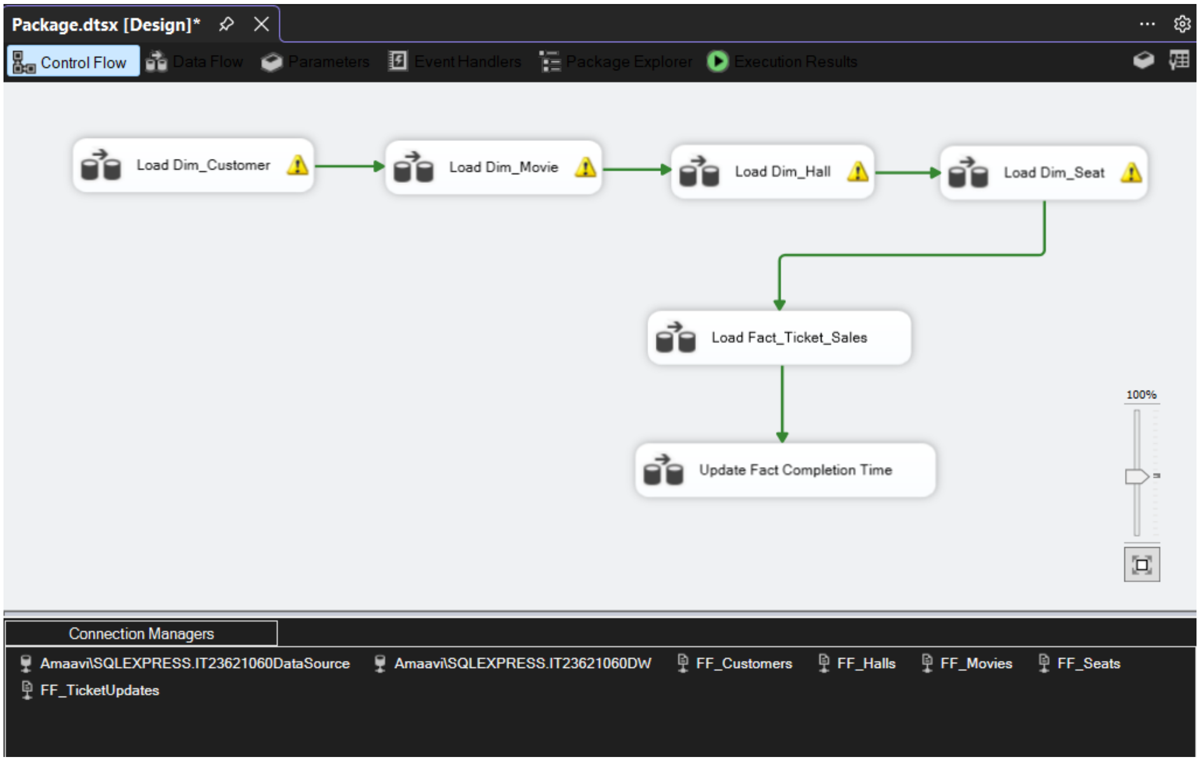
1. Loading dimension table
    - Each dimension table was loaded from flat files (CSV) using SSIS data flow tasks.
2. Loading fact table
    - The fact table was loaded from the ticket_source table in SQL Server.
    - Lookup transformations were used to replace business keys with surrogate keys from dimension tables, ensuring referential integrity in the data warehouse.

The SSIS package was designed using multiple Data Flow tasks, each responsible for loading a specific table.

Dimension tables were loaded before the fact table to ensure that surrogate keys were available for lookup transformation.

The Control Flow includes:

- Load_Dim_Customer
- Load_Dim_Movie
- Load_Dim_Hall
- Load_Dim_Seat
- Load_Fact_Ticket_Sales
- Update Fact Completion Time



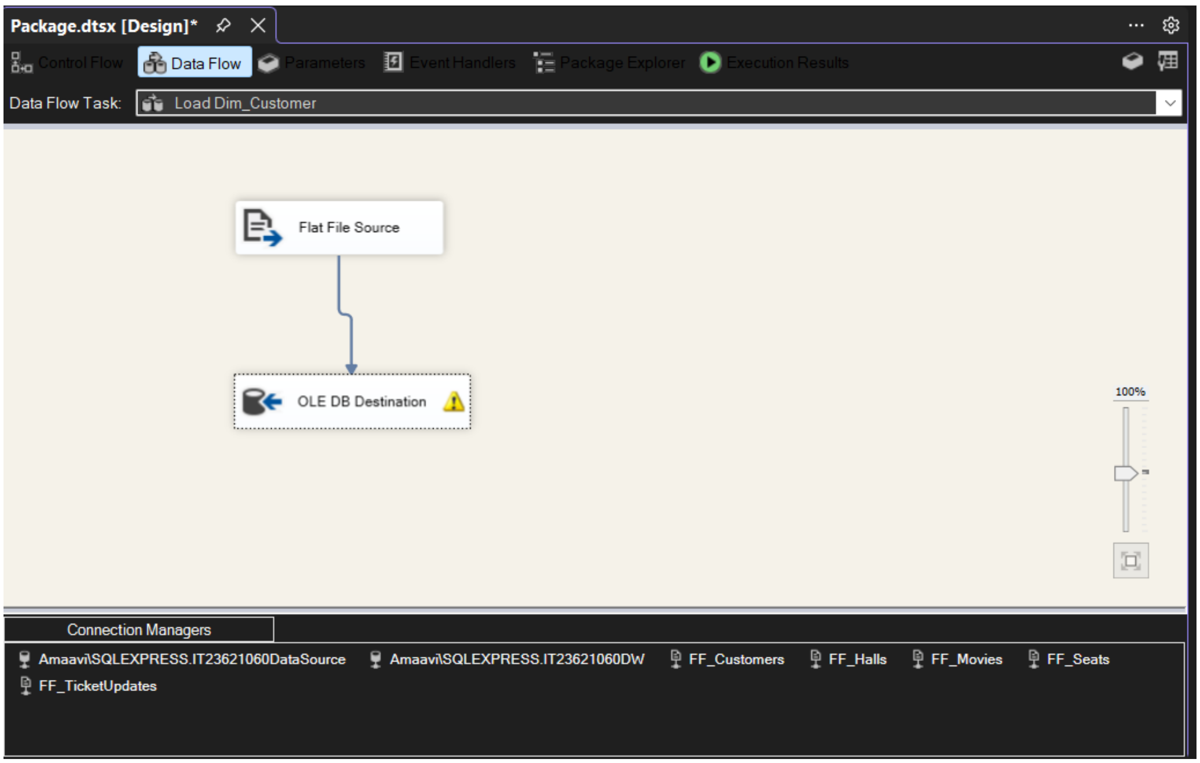
SSIS Control Flow for ETL Process

1. Load Dim_Customer

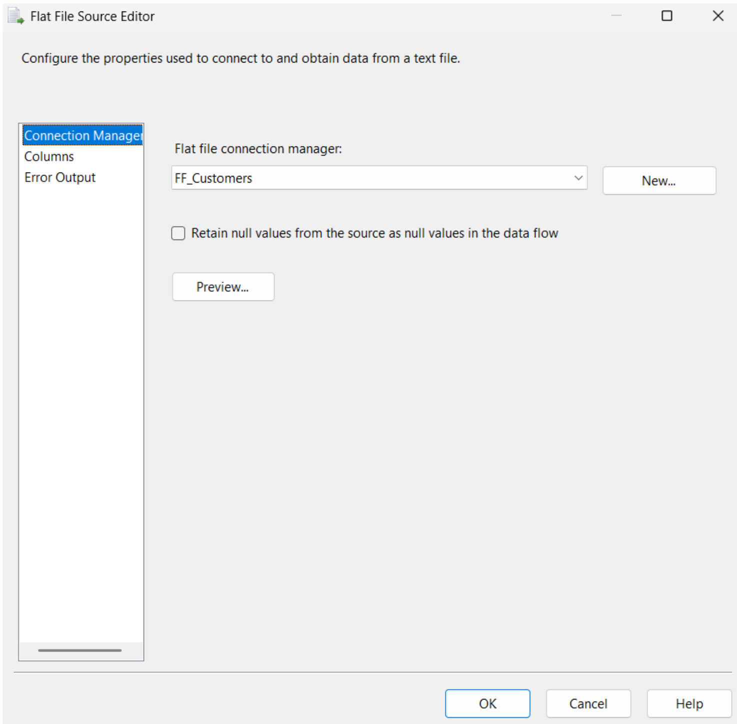
Source: Customer.csv

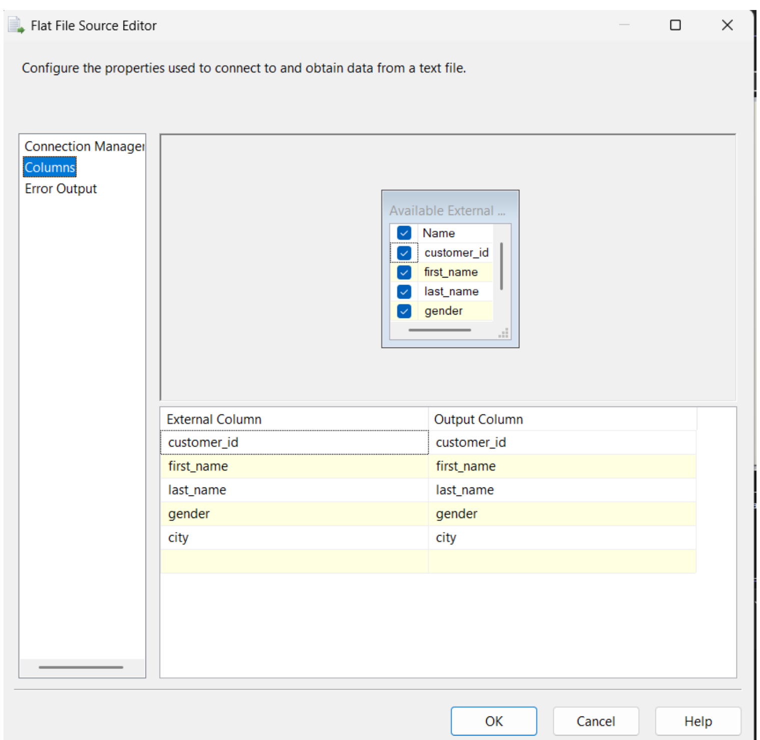
Component: Flat file Source

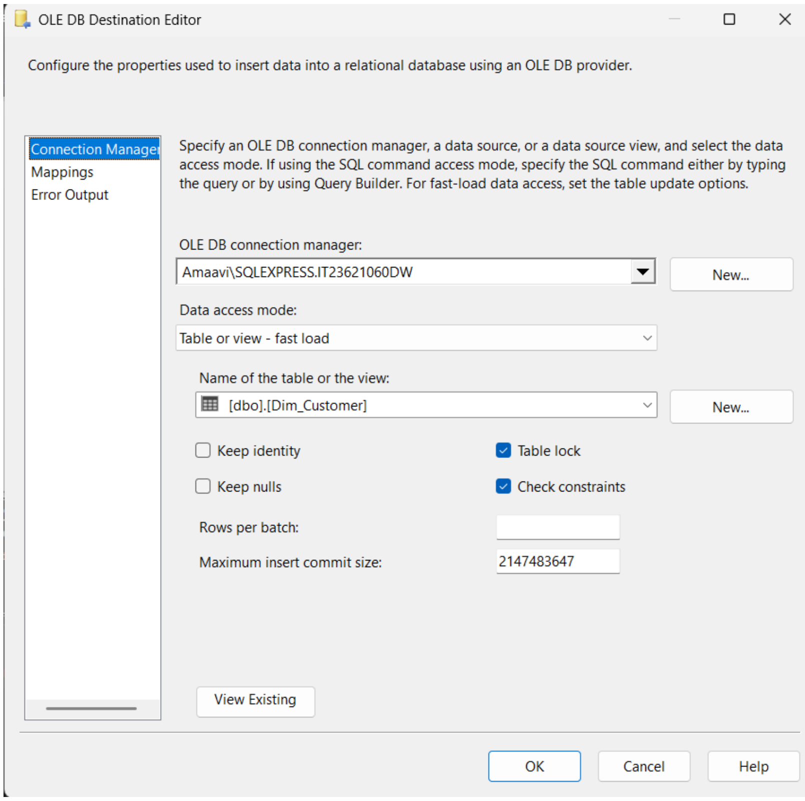
Data loaded into: Dim_Customer using OLE DB Destination

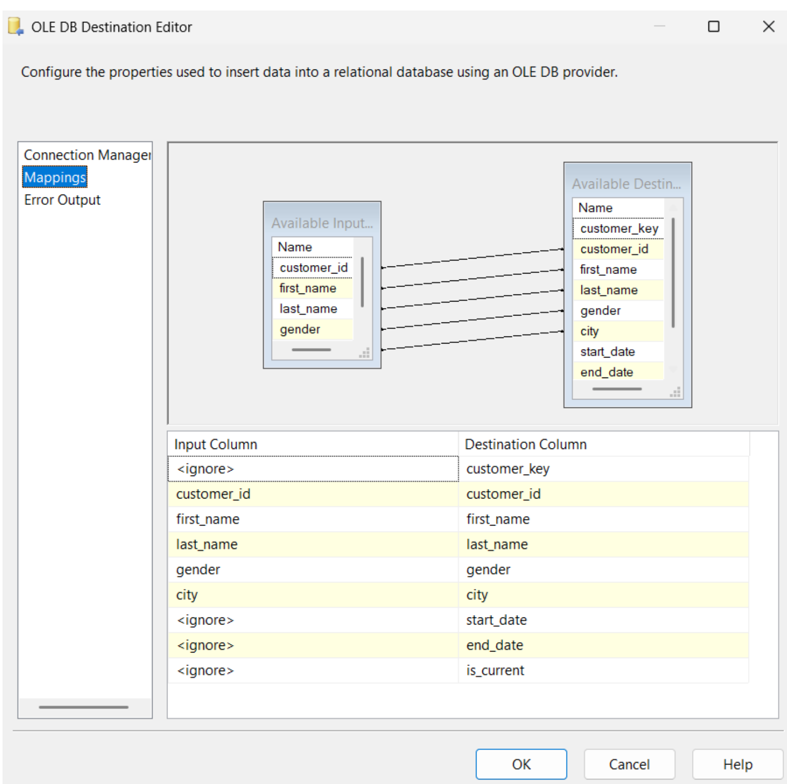


Data Flow for Loading Customer Dimension


Flat File Source Editor
Configure the properties used to connect to and obtain data from a text file.
Connection Manager
Columns
Error Output
Flat file connection manager:
FF_Customers
New...
Retain null values from the source as null values in the data flow
Preview...
OK
Cancel
Help


Flat File Source Editor
Configure the properties used to connect to and obtain data from a text file.
Connection Manager
Columns
Error Output
Available External ...
Name
customer_id
first_name
last_name
gender
External Column
Output Column
customer_id
customer_id
first_name
first_name
last_name
last_name
gender
gender
city
city
OK
Cancel
Help

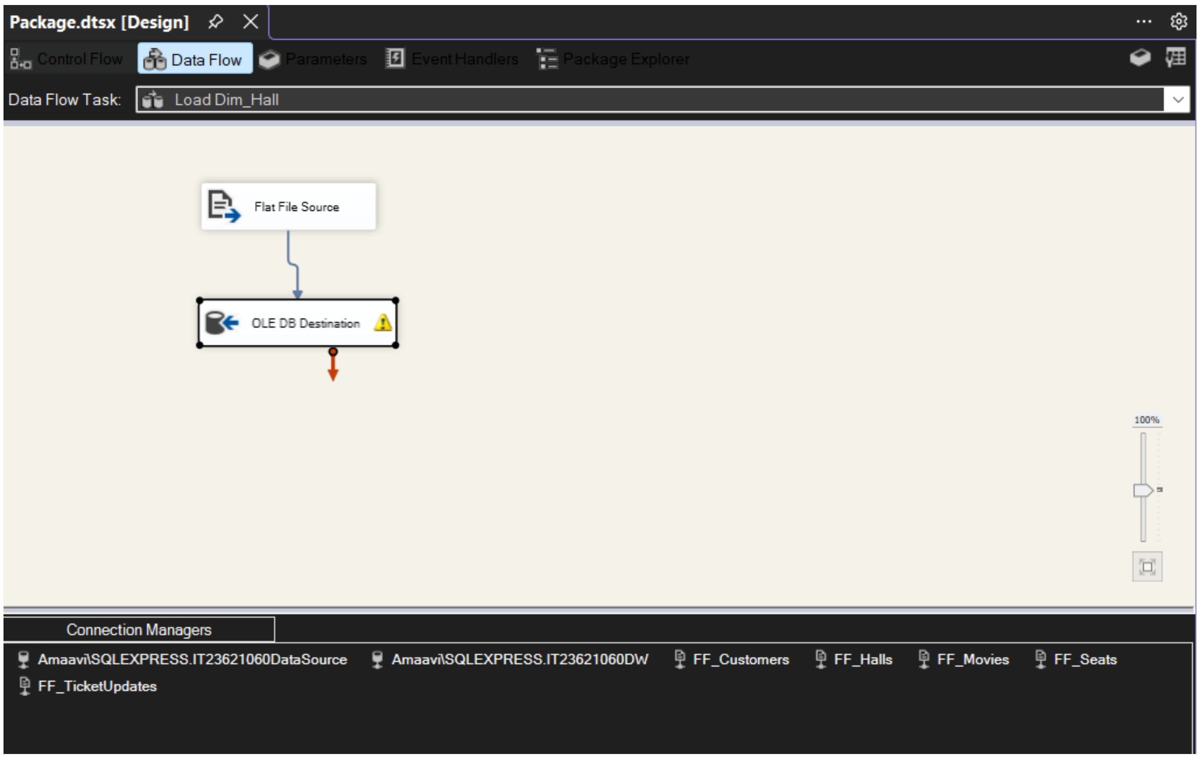

OLE DB Destination Editor
Configure the properties used to insert data into a relational database using an OLE DB provider.
Connection Manager
Mappings
Error Output
Specify an OLE DB connection manager, a data source, or a data source view, and select the data access mode. If using the SQL command access mode, specify the SQL command either by typing the query or by using Query Builder. For fast-load data access, set the table update options.
OLE DB connection manager:
Amaavi\SQLEXPRESS.IT23621060DW
New...
Data access mode:
Table or view - fast load
Name of the table or the view:
[dbo].[Dim_Customer]
New...
Keep identity
Table lock
Keep nulls
Check constraints
Rows per batch:
Maximum insert commit size:
2147483647
View Existing
OK
Cancel
Help


OLE DB Destination Editor
Configure the properties used to insert data into a relational database using an OLE DB provider.
Connection Manager
Mappings
Error Output
Available Input...
Name
customer_id
first_name
last_name
gender
Available Destin...
Name
customer_key
customer_id
first_name
last_name
gender
city
start_date
end_date
Input Column
Destination Column
<ignore>
customer_key
customer_id
customer_id
first_name
first_name
last_name
last_name
gender
gender
city
city
<ignore>
start_date
<ignore>
end_date
<ignore>
is_current
OK
Cancel
Help

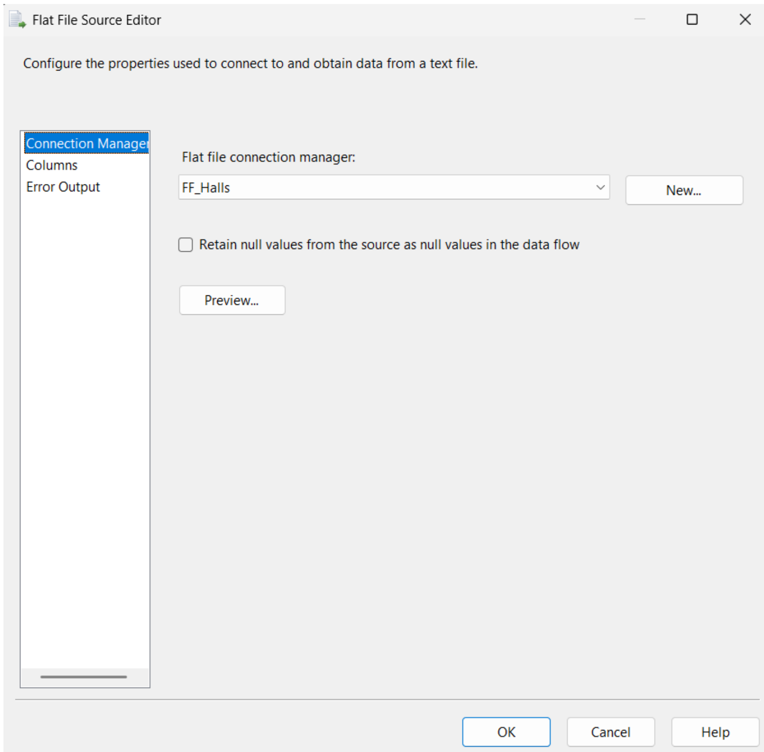
2. Load Dim_Hall

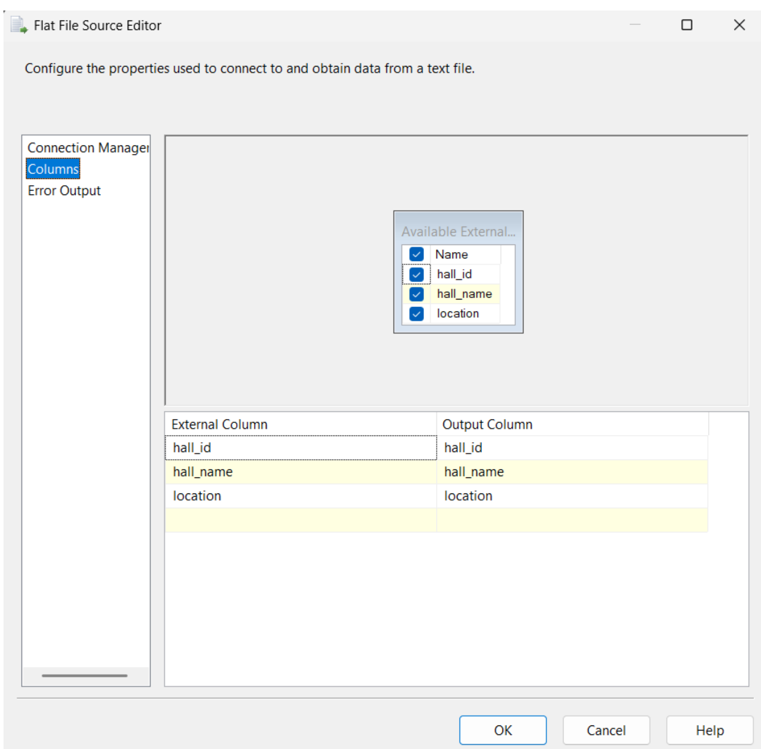
Source: Halls.csv

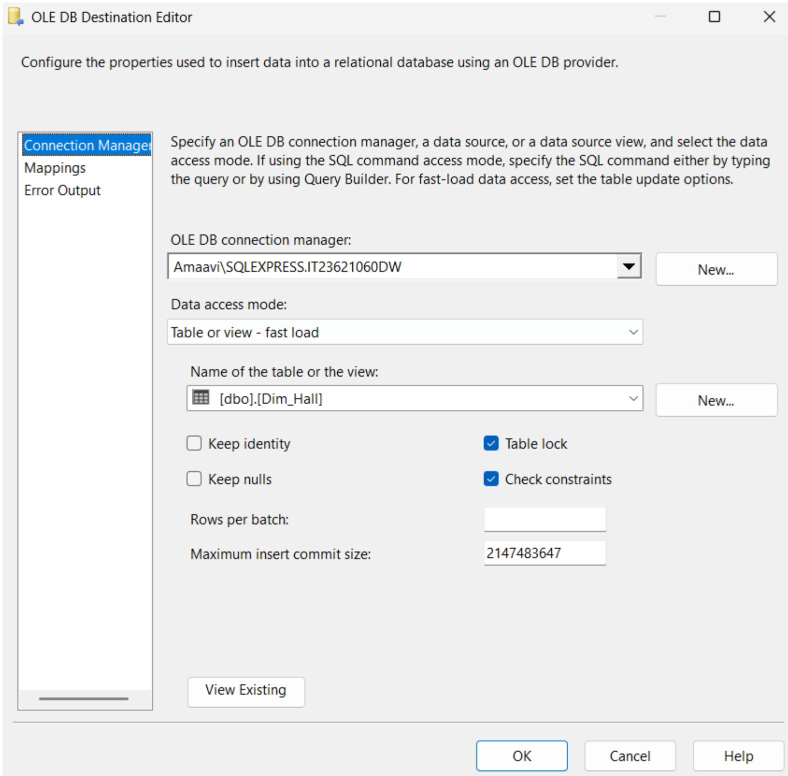
Component: Flat file Source

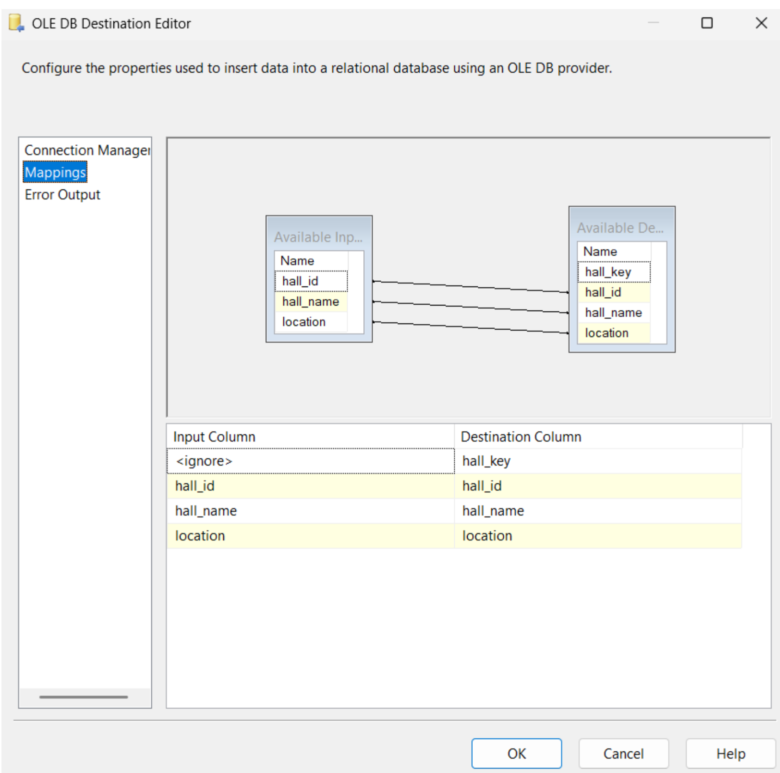


Data Flow for Loading Hall Dimension

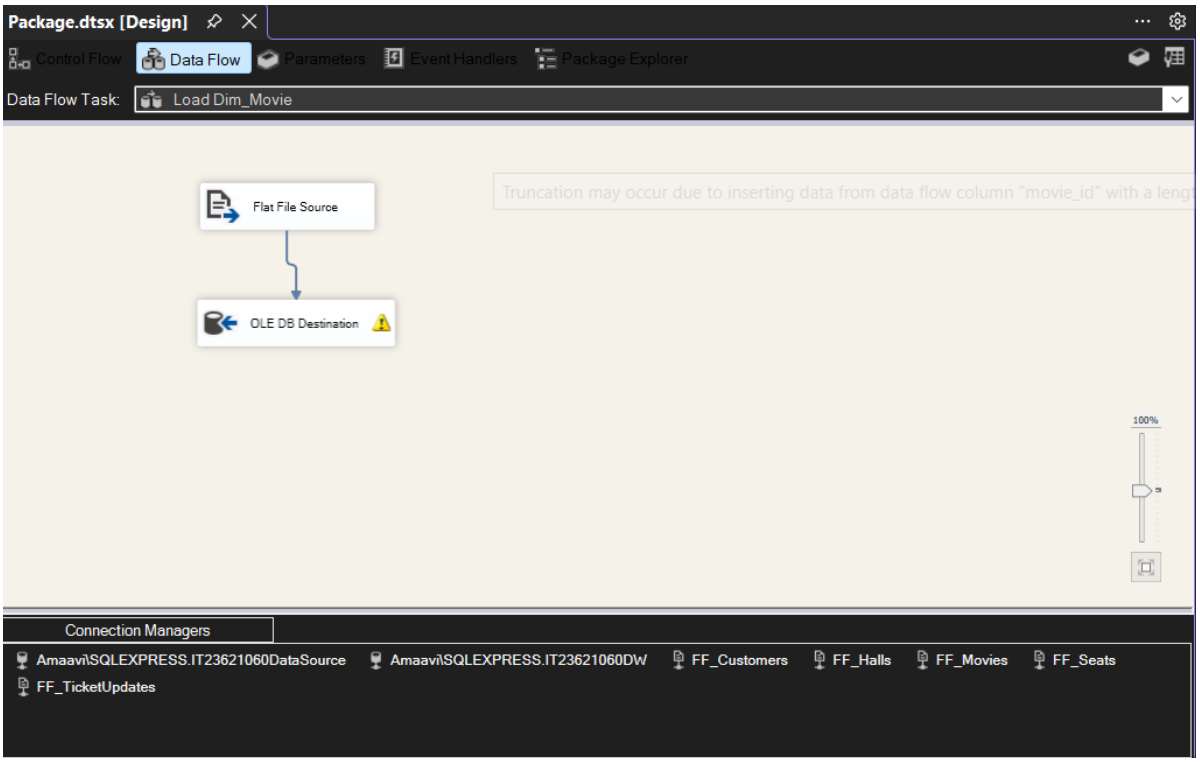

Flat File Source Editor
Configure the properties used to connect to and obtain data from a text file.
Connection Manager
Columns
Error Output
Flat file connection manager:
FF_Halls
New...
Retain null values from the source as null values in the data flow
Preview...
OK
Cancel
Help


Flat File Source Editor
Configure the properties used to connect to and obtain data from a text file.
Connection Manager
Columns
Error Output
Available External...
Name
hall_id
hall_name
location
External Column
Output Column
hall_id
hall_id
hall_name
hall_name
location
location
OK
Cancel
Help

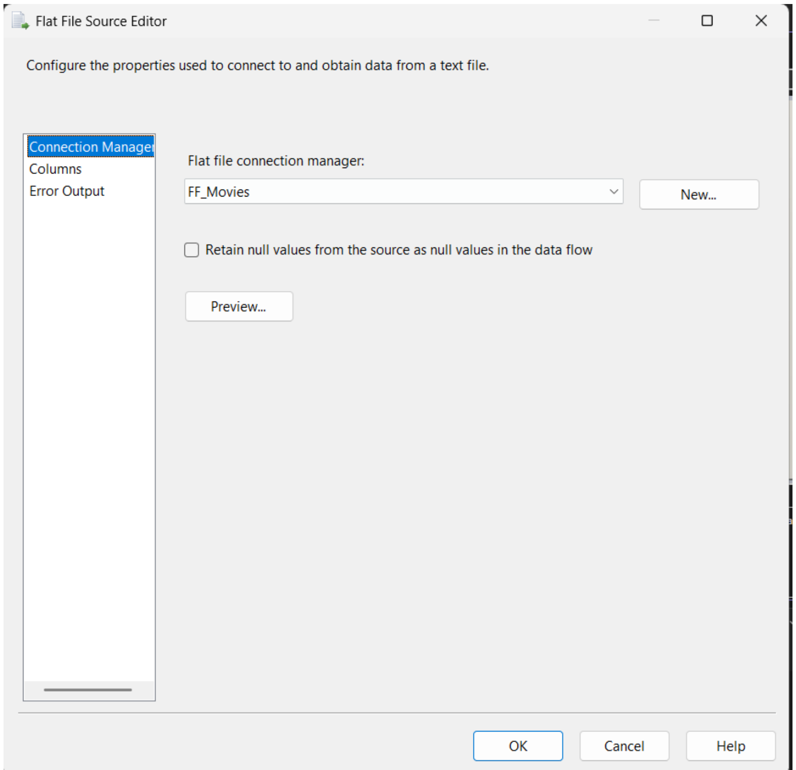

OLE DB Destination Editor
Configure the properties used to insert data into a relational database using an OLE DB provider.
Connection Manager
Mappings
Error Output
Specify an OLE DB connection manager, a data source, or a data source view, and select the data access mode. If using the SQL command access mode, specify the SQL command either by typing the query or by using Query Builder. For fast-load data access, set the table update options.
OLE DB connection manager:
Amaavi\SQLEXPRESS.IT23621060DW
New...
Data access mode:
Table or view - fast load
Name of the table or the view:
[dbo].[Dim_Hall]
New...
Keep identity
Table lock
Keep nulls
Check constraints
Rows per batch:
Maximum insert commit size:
2147483647
View Existing
OK
Cancel
Help

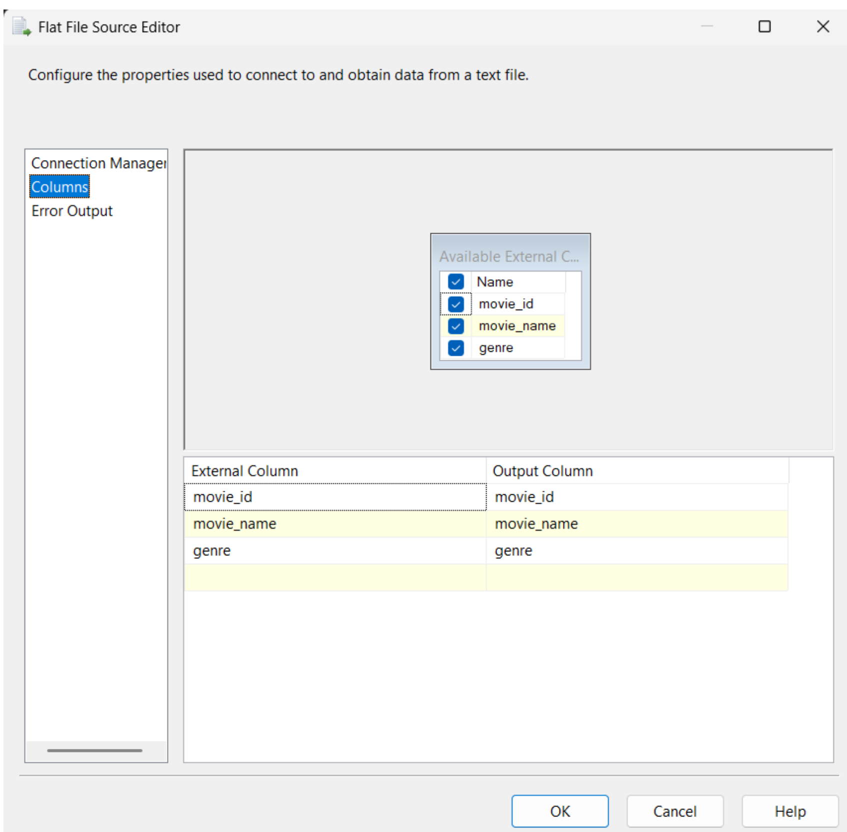

OLE DB Destination Editor
Configure the properties used to insert data into a relational database using an OLE DB provider.
Connection Manager
Mappings
Error Output
Available Inp...
Name
hall_id
hall_name
location
Available De...
Name
hall_key
hall_id
hall_name
location
Input Column
Destination Column
<ignore>
hall_key
hall_id
hall_id
hall_name
hall_name
location
location
OK
Cancel
Help

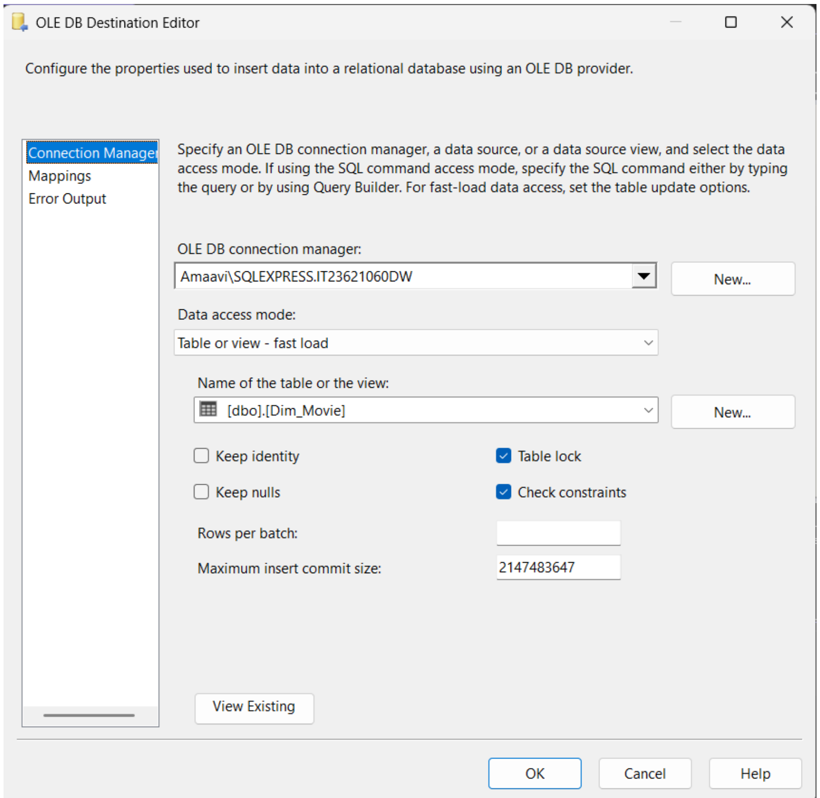
3. Load Dim_Movie

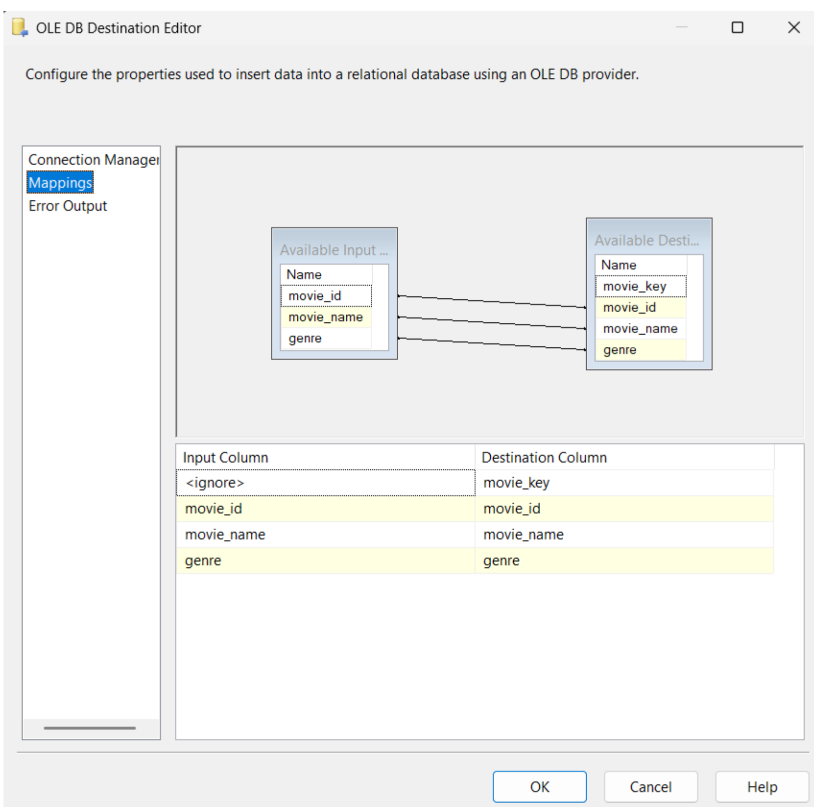


Data Flow for Loading Movie Dimension

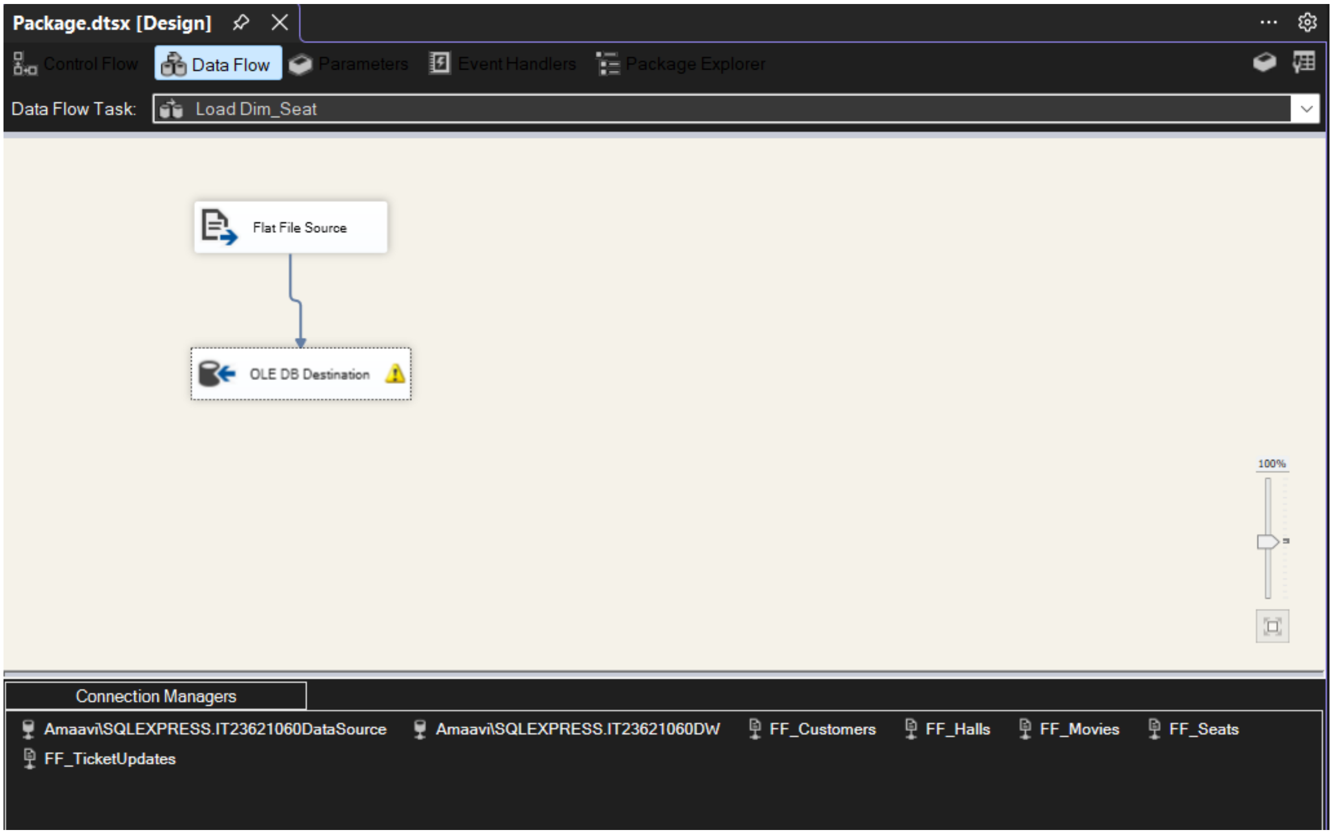

Flat File Source Editor
Configure the properties used to connect to and obtain data from a text file.
Connection Manager
Columns
Error Output
Flat file connection manager:
FF_Movies
New...
Retain null values from the source as null values in the data flow
Preview...
OK
Cancel
Help


Flat File Source Editor
Configure the properties used to connect to and obtain data from a text file.
Connection Manager
Columns
Error Output
Available External C...
Name
movie_id
movie_name
genre
External Column
Output Column
movie_id
movie_id
movie_name
movie_name
genre
genre
OK
Cancel
Help

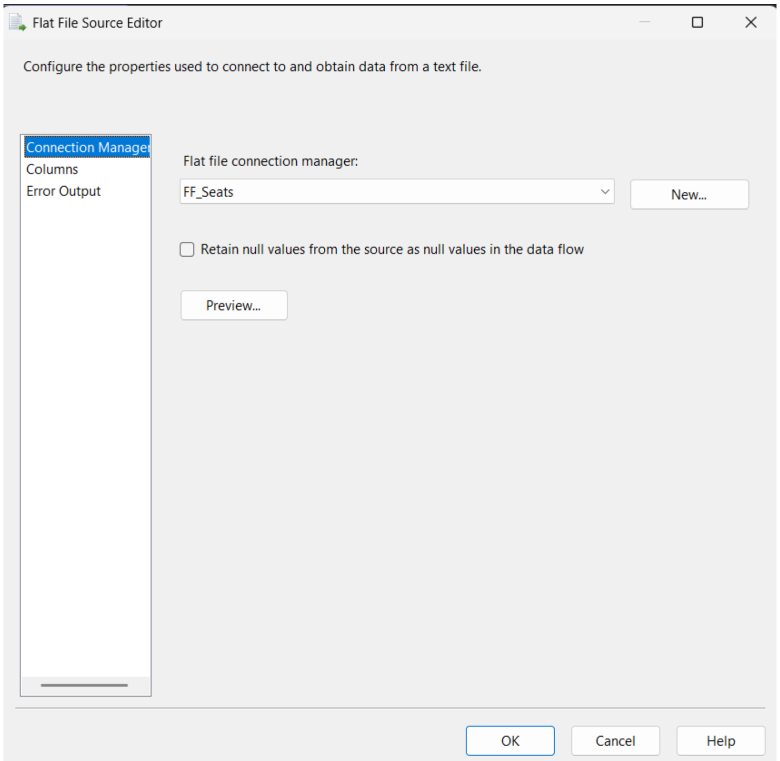

OLE DB Destination Editor
Configure the properties used to insert data into a relational database using an OLE DB provider.
Connection Manager
Mappings
Error Output
Specify an OLE DB connection manager, a data source, or a data source view, and select the data access mode. If using the SQL command access mode, specify the SQL command either by typing the query or by using Query Builder. For fast-load data access, set the table update options.
OLE DB connection manager:
Amaavi\SQLEXPRESS.IT23621060DW
New...
Data access mode:
Table or view - fast load
Name of the table or the view:
[dbo].[Dim_Movie]
New...
Keep identity
Table lock
Keep nulls
Check constraints
Rows per batch:
Maximum insert commit size:
2147483647
View Existing
OK
Cancel
Help

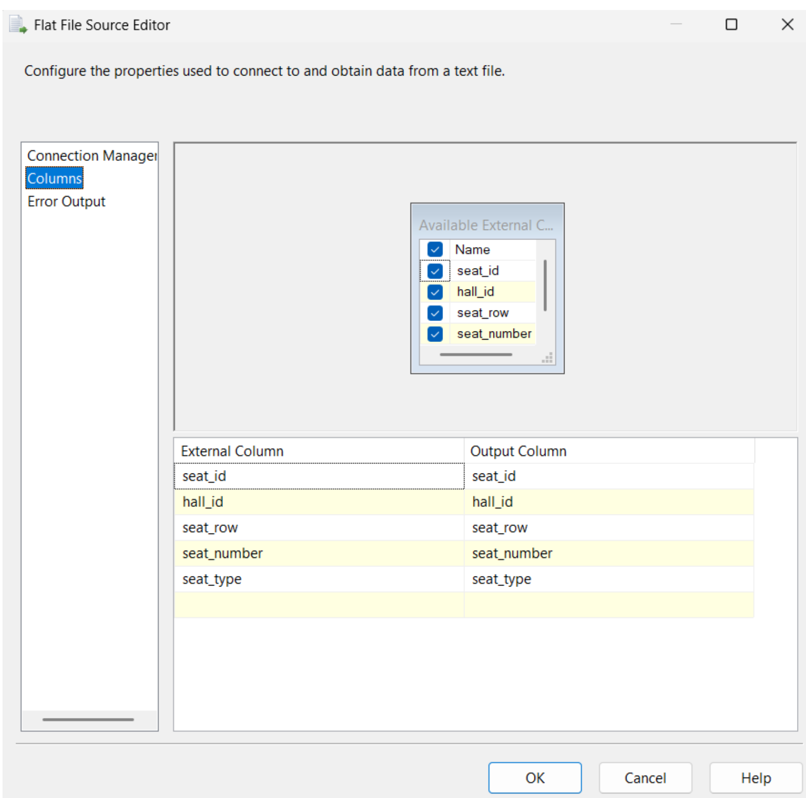

OLE DB Destination Editor
Configure the properties used to insert data into a relational database using an OLE DB provider.
Connection Manager
Mappings
Error Output
Available Input ...
Name
movie_id
movie_name
genre
Available Desti...
Name
movie_key
movie_id
movie_name
genre
Input Column
Destination Column
<ignore>
movie_key
movie_id
movie_id
movie_name
movie_name
genre
genre
OK
Cancel
Help

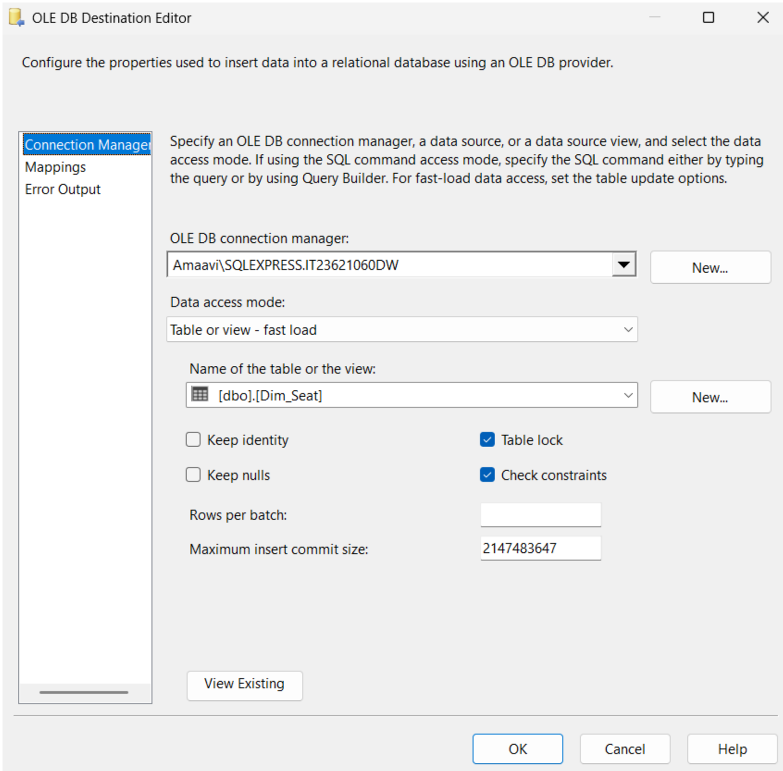
4. Load Dim_Seat

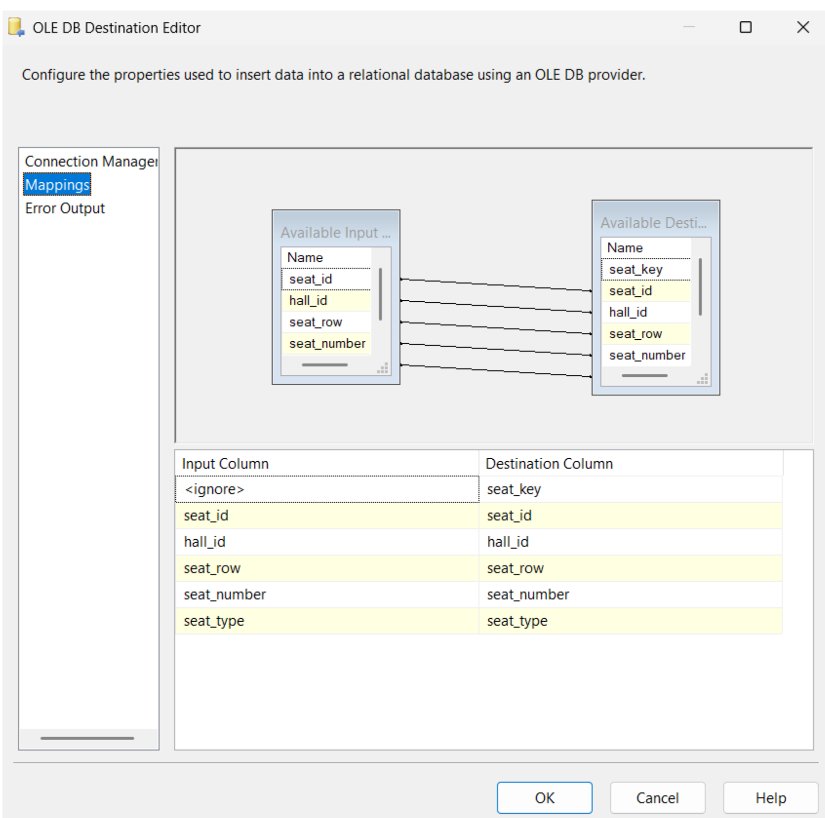


Data Flow for Loading Seat Dimension


Flat File Source Editor
Configure the properties used to connect to and obtain data from a text file.
Connection Manager
Columns
Error Output
Flat file connection manager:
FF_Seats
New...
Retain null values from the source as null values in the data flow
Preview...
OK
Cancel
Help

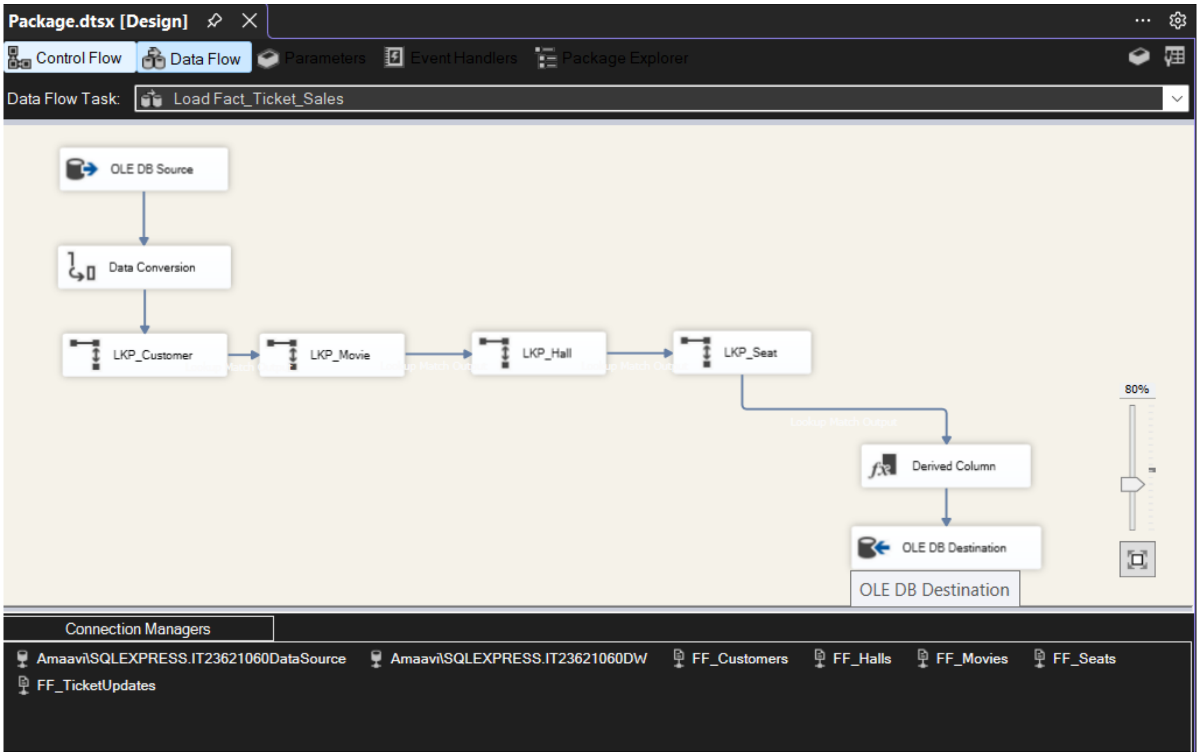

Flat File Source Editor
Configure the properties used to connect to and obtain data from a text file.
Connection Manager
Columns
Error Output
Available External C...
Name
seat_id
hall_id
seat_row
seat_number
External Column
Output Column
seat_id
seat_id
hall_id
hall_id
seat_row
seat_row
seat_number
seat_number
seat_type
seat_type
OK
Cancel
Help


OLE DB Destination Editor
Configure the properties used to insert data into a relational database using an OLE DB provider.
Connection Manager
Mappings
Error Output
Specify an OLE DB connection manager, a data source, or a data source view, and select the data access mode. If using the SQL command access mode, specify the SQL command either by typing the query or by using Query Builder. For fast-load data access, set the table update options.
OLE DB connection manager:
Amaavi\SQLEXPRESS.IT23621060DW
New...
Data access mode:
Table or view - fast load
Name of the table or the view:
[dbo].[Dim_Seat]
New...
Keep identity
Table lock
Keep nulls
Check constraints
Rows per batch:
Maximum insert commit size:
2147483647
View Existing
OK
Cancel
Help

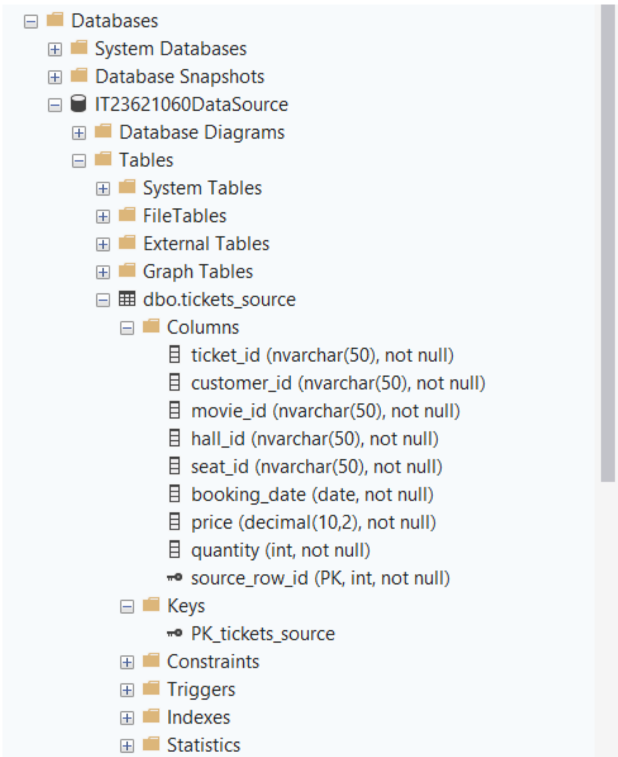

OLE DB Destination Editor
Configure the properties used to insert data into a relational database using an OLE DB provider.
Connection Manager
Mappings
Error Output
Available Input ...
Name
seat_id
hall_id
seat_row
seat_number
Available Desti...
Name
seat_key
seat_id
hall_id
seat_row
seat_number
Input Column
Destination Column
<ignore>
seat_key
seat_id
seat_id
hall_id
hall_id
seat_row
seat_row
seat_number
seat_number
seat_type
seat_type
OK
Cancel
Help

5. Fact Table Loading

The fact table was loaded using transactional data from the SQL Server source table Tickets_source.



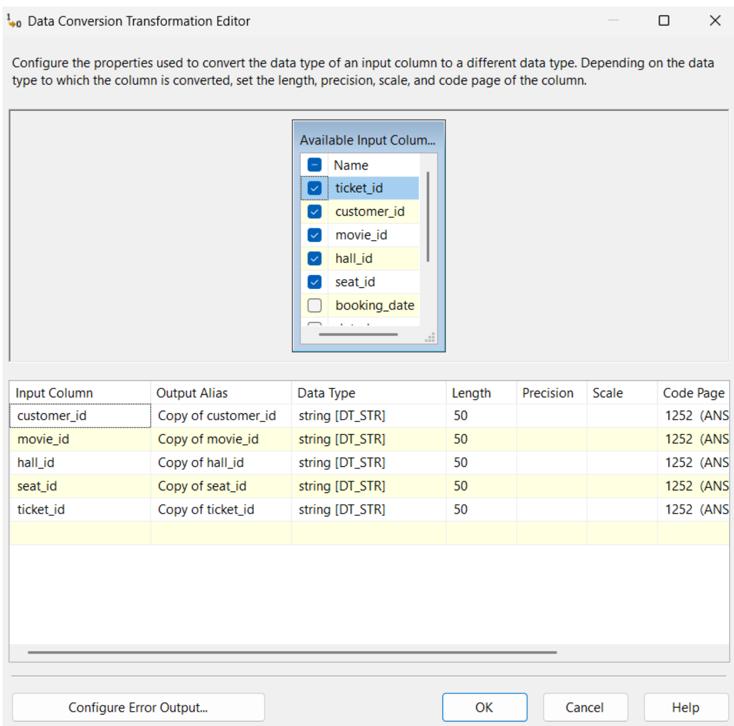
i. Data Extraction



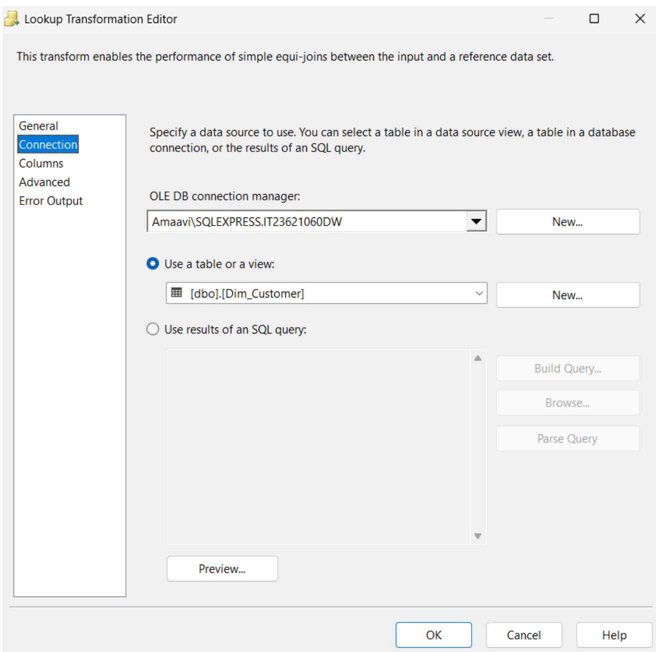
ii. Data Transformation

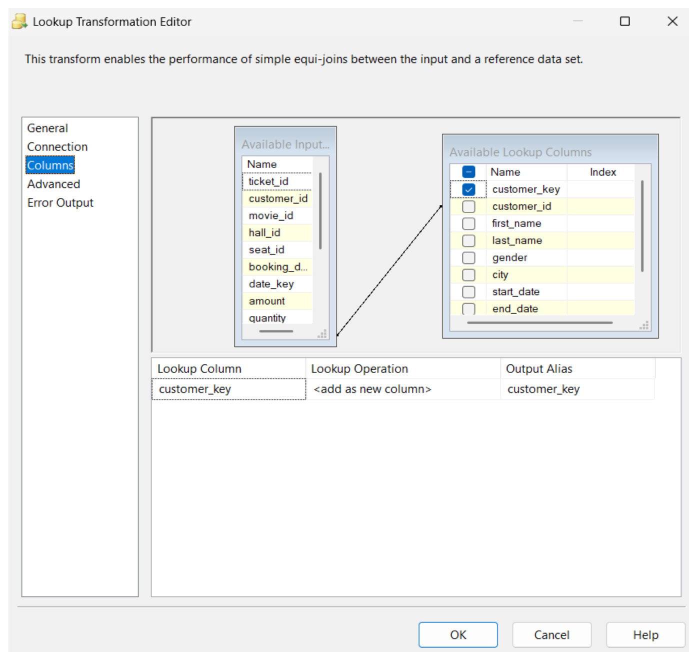
Several data transformation types are applied:

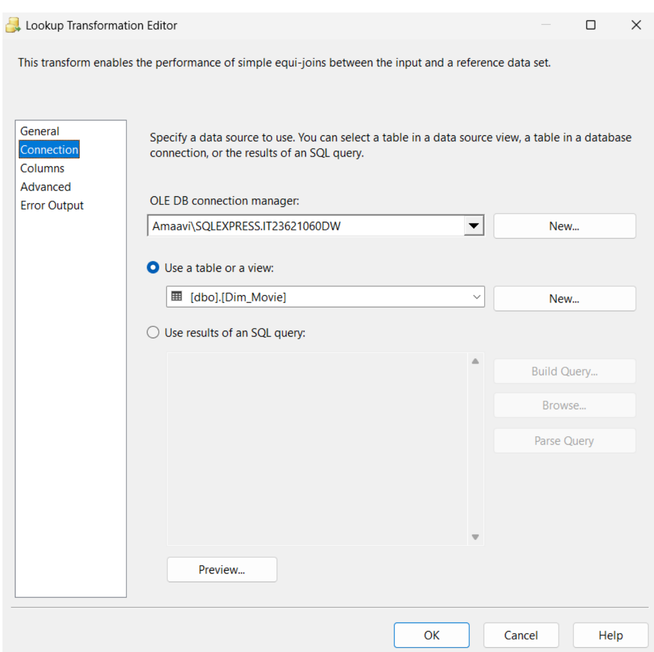
- Data conversion– Ensure data types match the destination table.

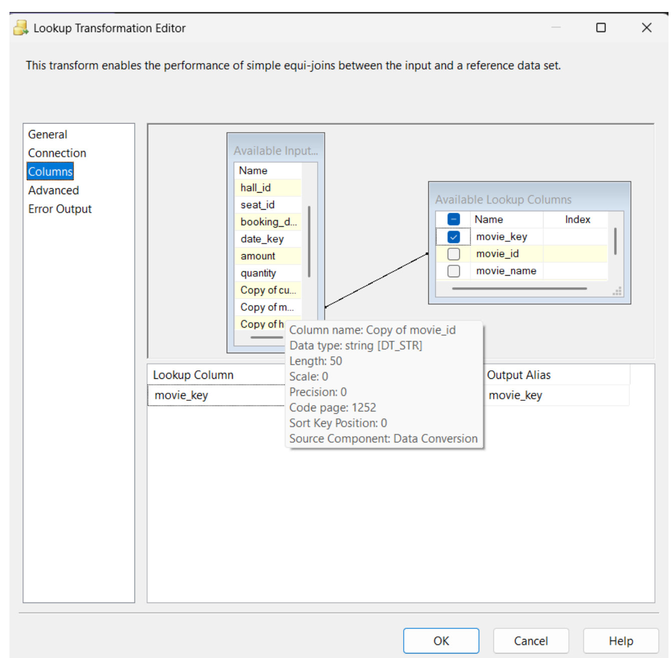


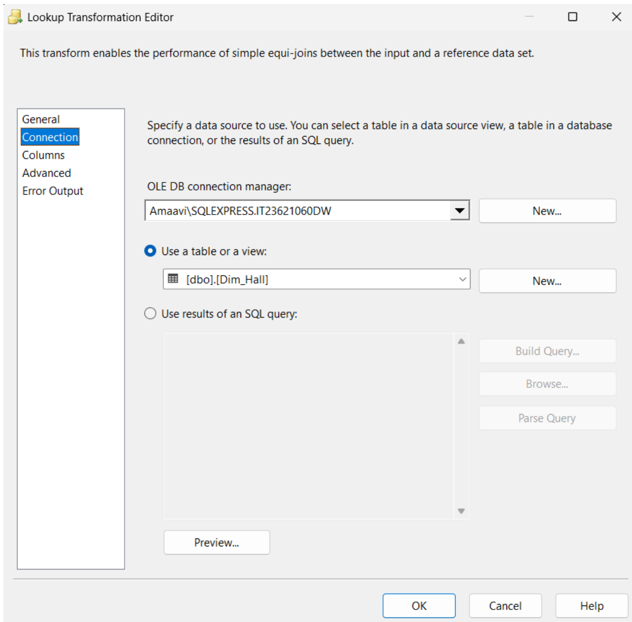
- Lookup Transformation- Used to retrieve surrogate keys from dimension tables.
  - Customer Lookup → customer_key
  - Movie Lookup → movie_key
  - Hall Lookup → hall_key
  - Seat Lookup → seat_key

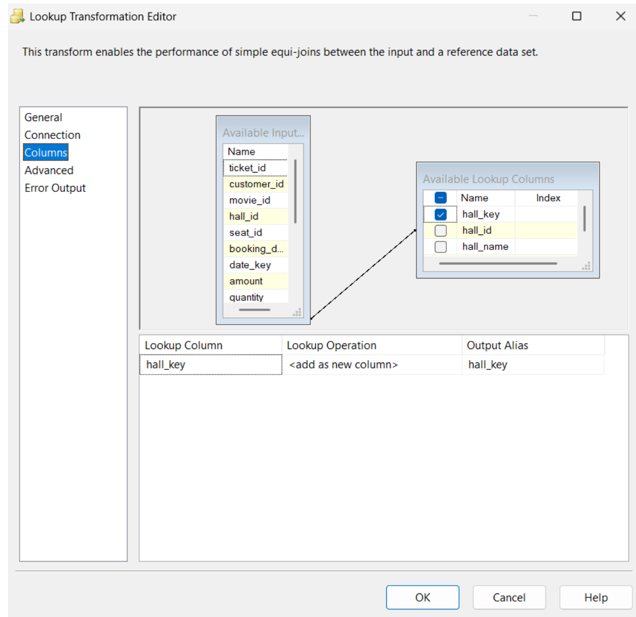


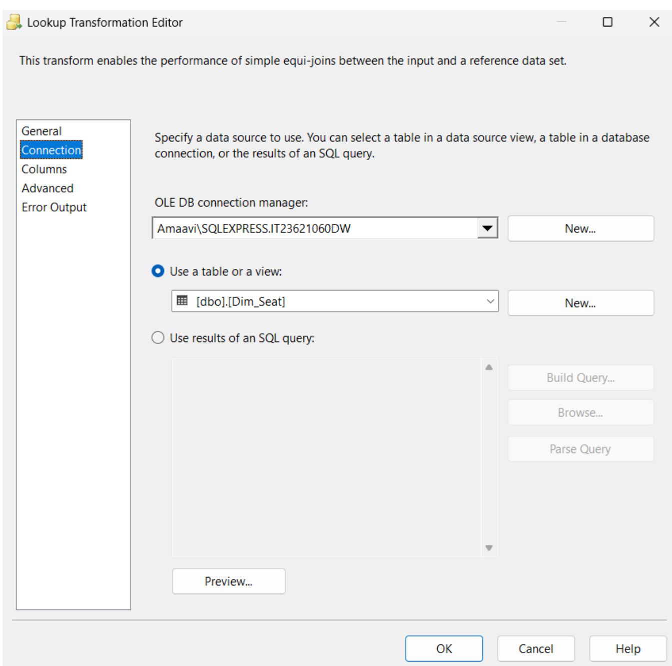


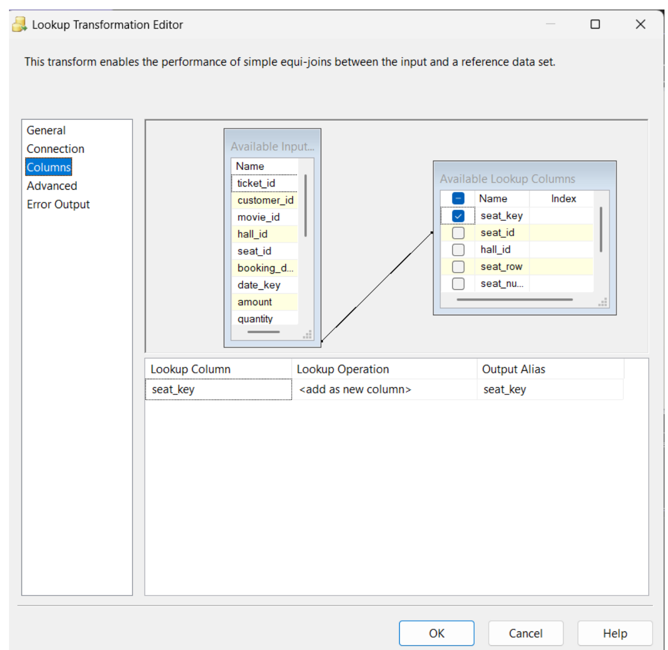




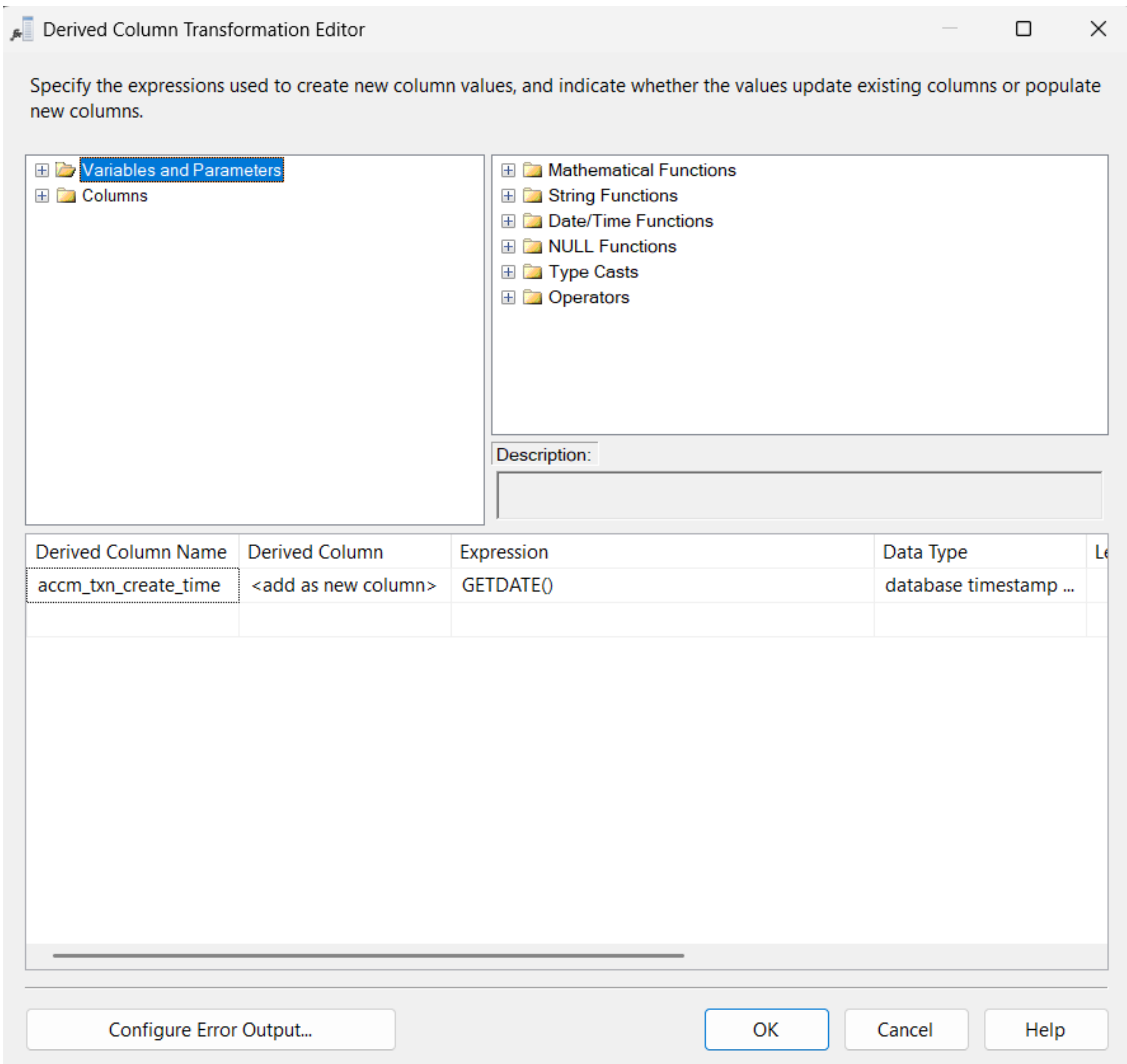

Lookup Transformation Editor
This transform enables the performance of simple equi-joins between the input and a reference data set.
General
Connection
Columns
Advanced
Error Output
Specify a data source to use. You can select a table in a data source view, a table in a database connection, or the results of an SQL query.
OLE DB connection manager:
Amaavi\SQLEXPRESS.IT23621060DW
New...
Use a table or a view:
[dbo].[Dim_Hall]
New...
Use results of an SQL query:
Build Query...
Browse...
Parse Query
Preview...
OK
Cancel
Help


Lookup Transformation Editor
This transform enables the performance of simple equi-joins between the input and a reference data set.
General
Connection
Columns
Advanced
Error Output
Available Input...
Name
ticket_id
customer_id
movie_id
hall_id
seat_id
booking_d...
date_key
amount
quantity
Available Lookup Columns
Name
Index
hall_key
hall_id
hall_name
Lookup Column
Lookup Operation
Output Alias
hall_key
<add as new column>
hall_key
OK
Cancel
Help

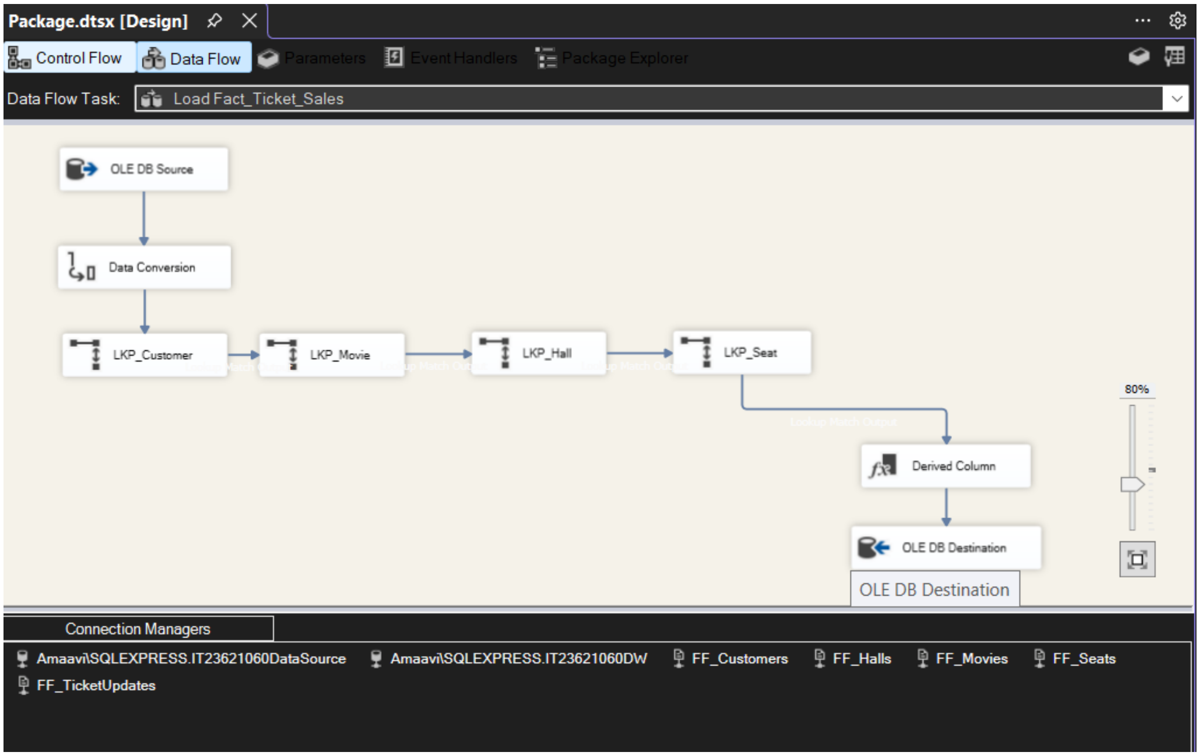

Lookup Transformation Editor
This transform enables the performance of simple equi-joins between the input and a reference data set.
General
Connection
Columns
Advanced
Error Output
Specify a data source to use. You can select a table in a data source view, a table in a database connection, or the results of an SQL query.
OLE DB connection manager:
Amaavi\SQLEXPRESS.IT23621060DW
New...
Use a table or a view:
[dbo].[Dim_Seat]
New...
Use results of an SQL query:
Build Query...
Browse...
Parse Query
Preview...
OK
Cancel
Help


Lookup Transformation Editor
This transform enables the performance of simple equi-joins between the input and a reference data set.
General
Connection
Columns
Advanced
Error Output
Available Input...
Name
ticket_id
customer_id
movie_id
hall_id
seat_id
booking_d...
date_key
amount
quantity
Available Lookup Columns
Name
Index
seat_key
seat_id
hall_id
seat_row
seat_nu...
Lookup Column
Lookup Operation
Output Alias
seat_key
<add as new column>
seat_key
OK
Cancel
Help

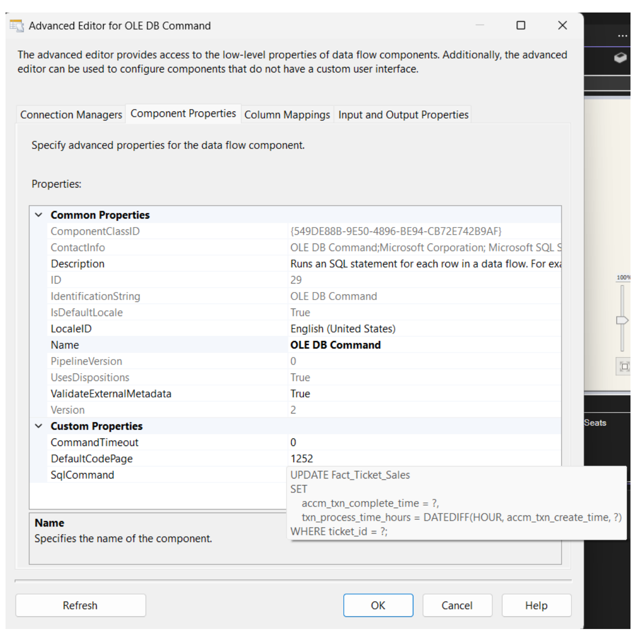
- Derived Column- Used to calculate or adjust fields such as,
    - accm_txn_create_time
    - date-related values



iii. Data Loading



Accumulating Fact Table Update- A separate ETL process was created to update the fact table using an additional CSV file.



Slowly Changing Dimension

The customer dimension was implemented as a Slowly Changing Dimension Type 2 (SCD Type 2) to maintain historical changes in customer data.

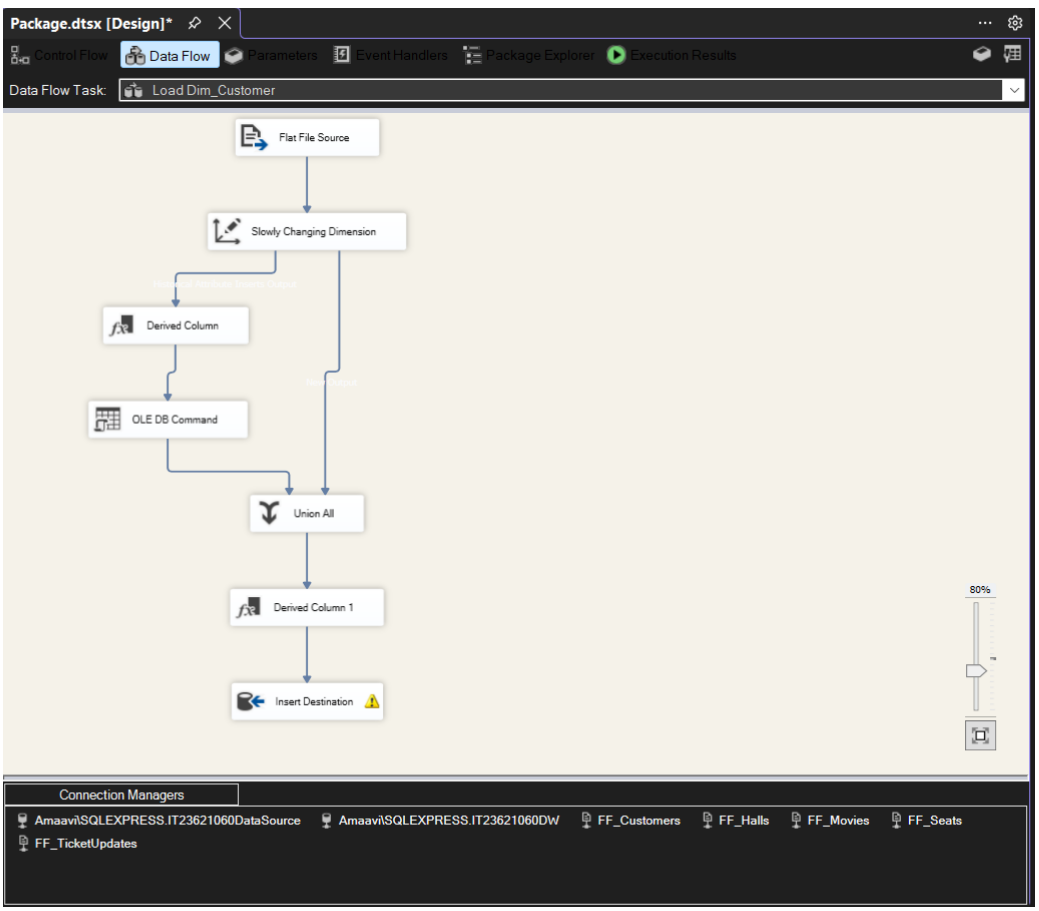
In this approach, when a customer's attribute (such as city or name) changes, the existing record is not overwritten. Instead, the existing record is marked as expired, and a new record is inserted with updated values.

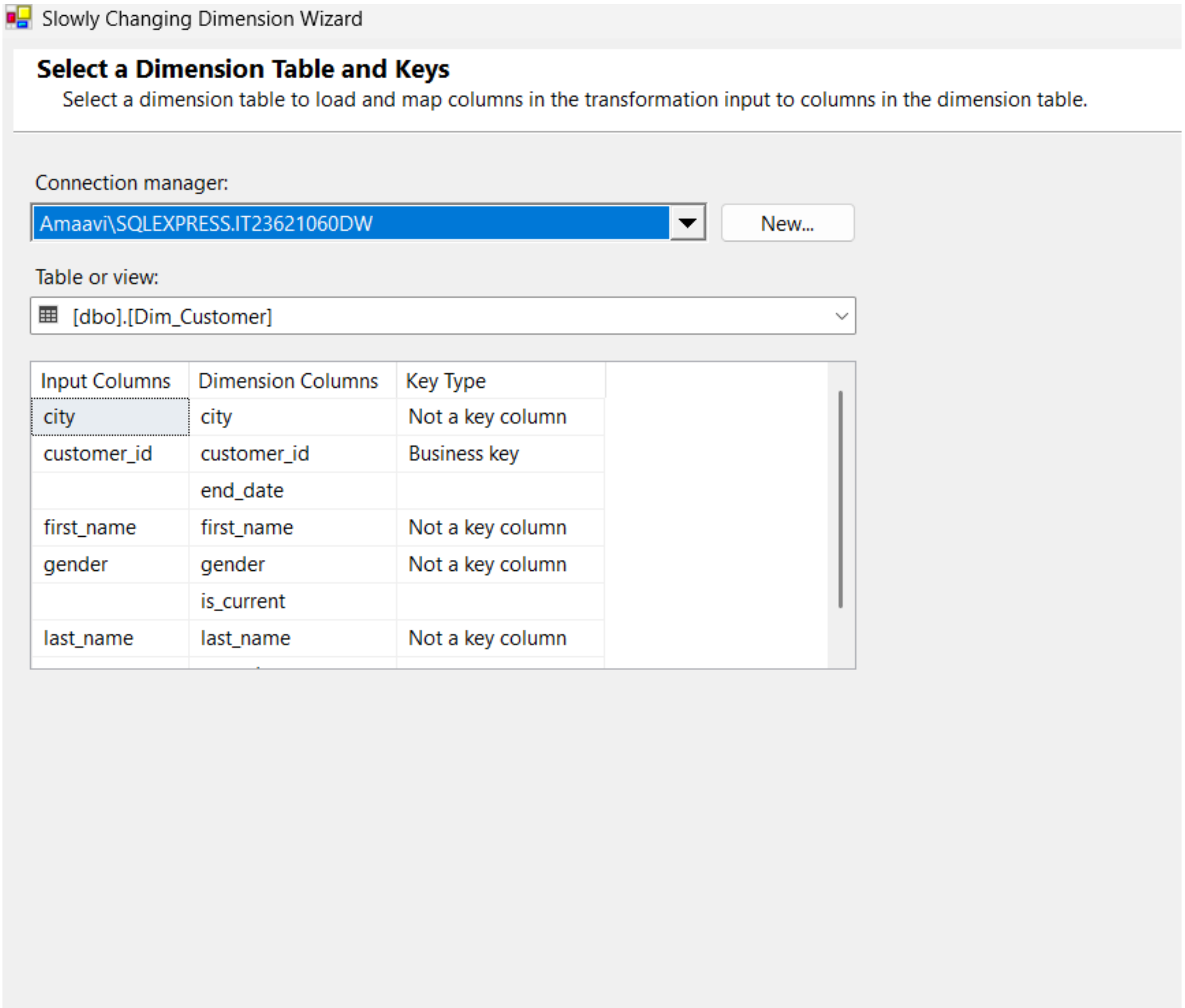
The following columns were used to support SCD Type 2:

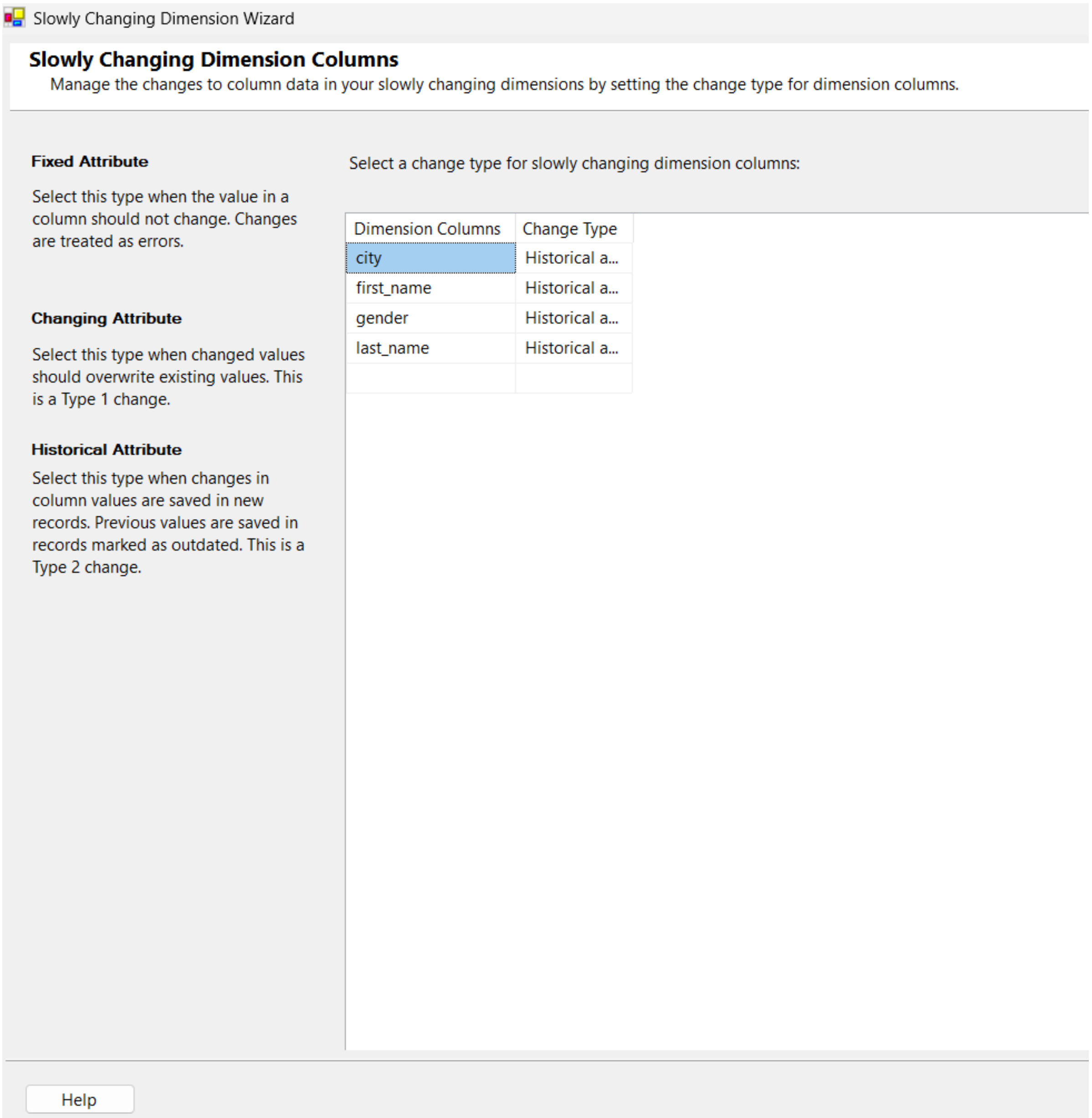
- start_date – indicates when the record became active
- end_date – indicates when the record expired
- is_current – identifies the current active record (1 = current, 0 = expired)

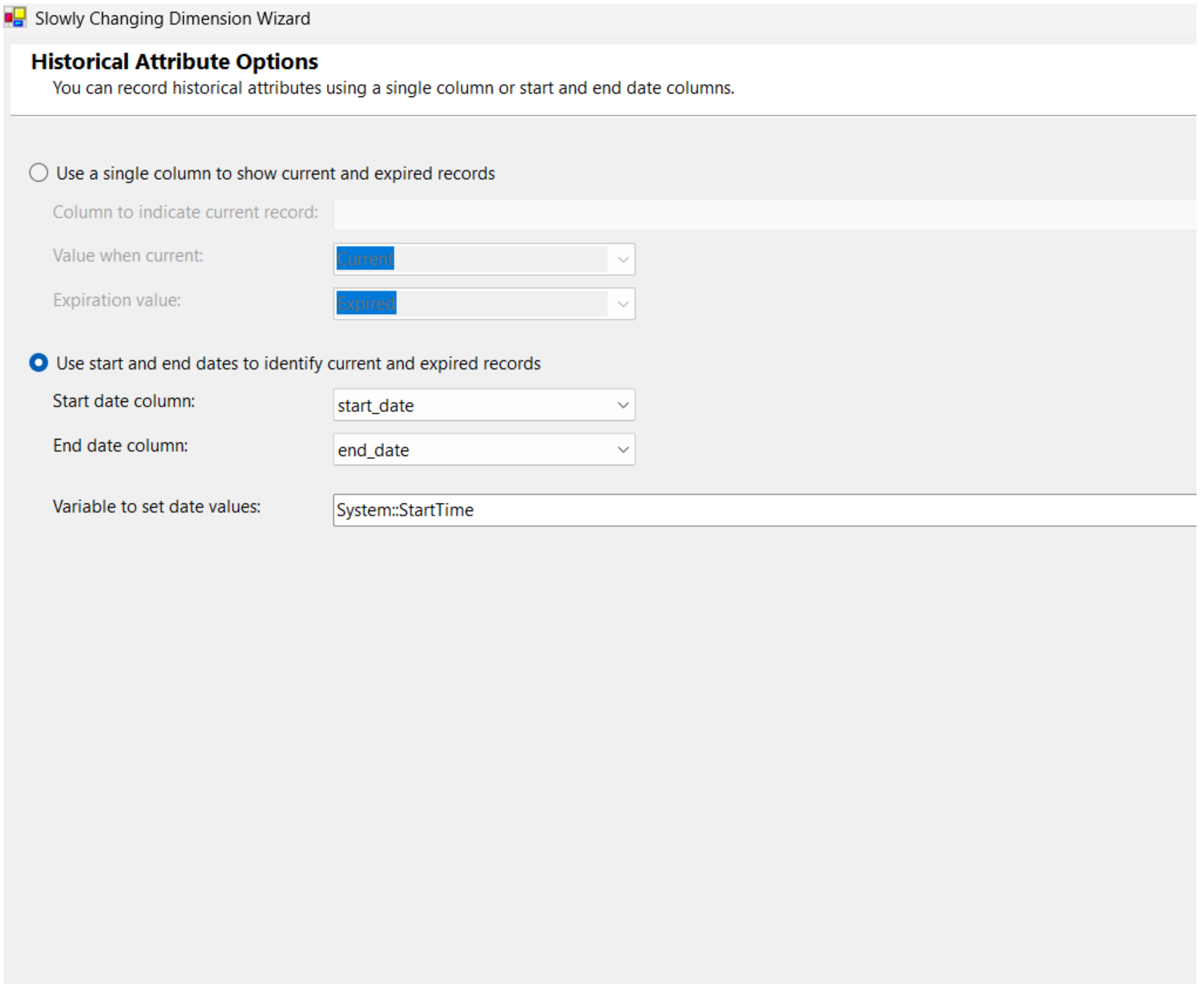
The SCD logic was implemented using SSIS transformations. When a change is detected:

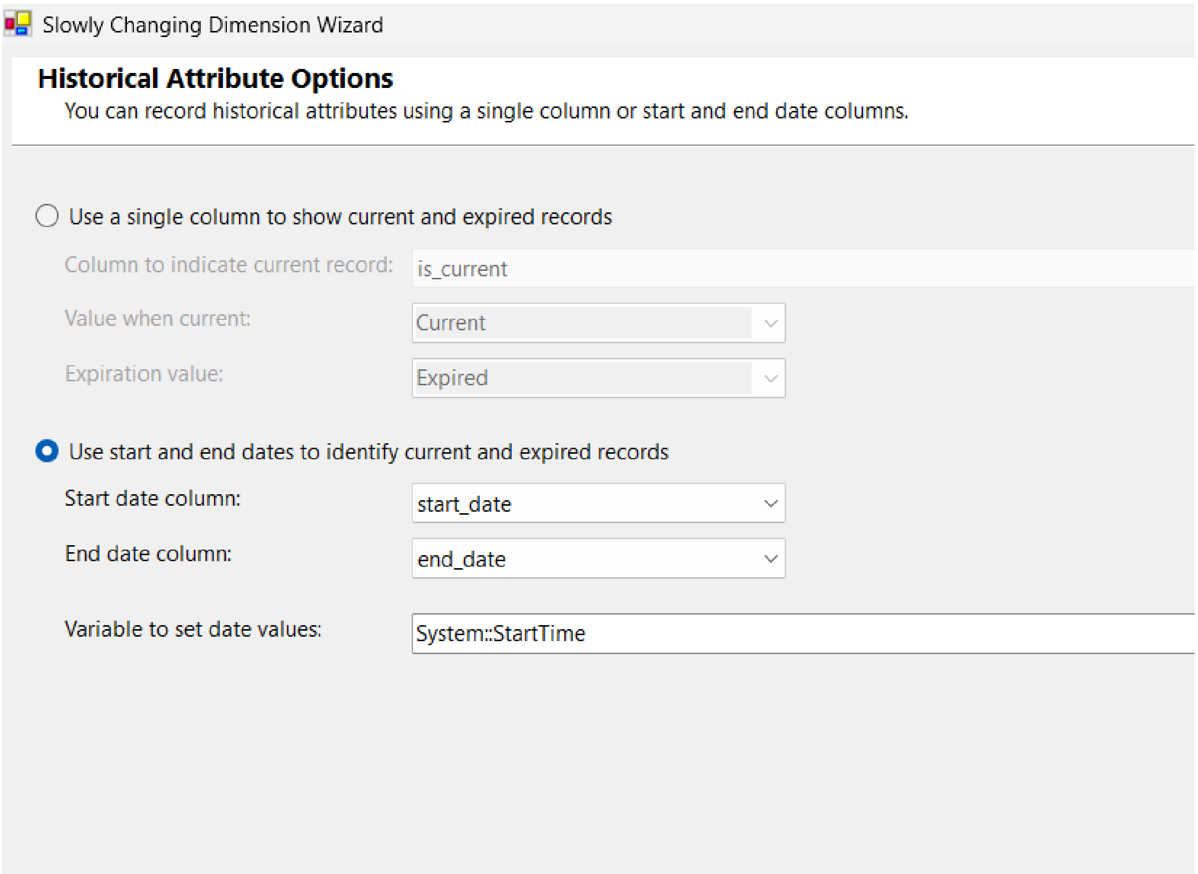
- The existing record is updated by setting end_date and is_current = 0
- A new record is inserted with updated values, start_date, and is_current = 1




Slowly Changing Dimension Wizard
Select a Dimension Table and Keys
Select a dimension table to load and map columns in the transformation input to columns in the dimension table.
Connection manager:
Amaavi\SQLEXPRESS.IT23621060DW
New...
Table or view:
[dbo].[Dim_Customer]
Input Columns
Dimension Columns
Key Type
city
city
Not a key column
customer_id
customer_id
Business key
end_date
first_name
first_name
Not a key column
gender
gender
Not a key column
is_current
last_name
last_name
Not a key column


Slowly Changing Dimension Wizard
Slowly Changing Dimension Columns
Manage the changes to column data in your slowly changing dimensions by setting the change type for dimension columns.
Fixed Attribute
Select this type when the value in a column should not change. Changes are treated as errors.
Changing Attribute
Select this type when changed values should overwrite existing values. This is a Type 1 change.
Historical Attribute
Select this type when changes in column values are saved in new records. Previous values are saved in records marked as outdated. This is a Type 2 change.
Select a change type for slowly changing dimension columns:
Dimension Columns
Change Type
city
Historical a...
first_name
Historical a...
gender
Historical a...
last_name
Historical a...
Help


Slowly Changing Dimension Wizard
Historical Attribute Options
You can record historical attributes using a single column or start and end date columns.
Use a single column to show current and expired records
Column to indicate current record:
Value when current:
Current
Expiration value:
Expired
Use start and end dates to identify current and expired records
Start date column:
start_date
End date column:
end_date
Variable to set date values:
System::StartTime


Slowly Changing Dimension Wizard
Historical Attribute Options
You can record historical attributes using a single column or start and end date columns.
Use a single column to show current and expired records
Column to indicate current record:
is_current
Value when current:
Current
Expiration value:
Expired
Use start and end dates to identify current and expired records
Start date column:
start_date
End date column:
end_date
Variable to set date values:
System::StartTime

```sql
select * from Dim_Customer;
```

100 %   No issues found

Results   Messages

	customer_key	customer_id	first_name	last_name	gender	city	start_date	end_date	is_current
1	1	C001	Amal	Perera	Male	Colombo	2026-03-28 13:28:15.000	NULL	1
2	2	C002	Nimali	Fernando	Female	Gampaha	2026-03-28 13:28:15.000	NULL	1
3	3	C003	Kamal	Silva	Male	Kandy	2026-03-28 13:28:15.000	NULL	1
4	4	C004	Sanduni	Jayasinghe	Female	Matara	2026-03-28 13:28:15.000	NULL	1
5	5	C005	Tharindu	Wijesinghe	Male	Negombo	2026-03-28 13:28:15.000	NULL	1
6	6	C006	Dilani	Perera	Female	Colombo	2026-03-28 13:28:15.000	NULL	1
7	7	C007	Kasun	Bandara	Male	Kurunegala	2026-03-28 13:28:15.000	NULL	1
8	8	C008	Shenali	Rodrigo	Female	Kalutara	2026-03-28 13:28:15.000	NULL	1
9	9	C009	Chathura	De Silva	Male	Galle	2026-03-28 13:28:15.000	NULL	1
10	10	C010	Imesha	Fernando	Female	Panadura	2026-03-28 13:28:15.000	NULL	1
11	11	C011	Ravindu	Perera	Male	Colombo	2026-03-28 13:28:15.000	NULL	1
12	12	C012	Thilini	Silva	Female	Kandy	2026-03-28 13:28:15.000	NULL	1
13	13	C013	Pasindu	Jayawardena	Male	Matara	2026-03-28 13:28:15.000	NULL	1
14	14	C014	Hiruni	Samarasinghe	Female	Gampaha	2026-03-28 13:28:15.000	NULL	1
15	15	C015	Isuru	Wickramasinghe	Male	Negombo	2026-03-28 13:28:15.000	NULL	1
16	16	C016	Sachini	Gunasekara	Female	Kegalle	2026-03-28 13:28:15.000	NULL	1
17	17	C017	Nuwan	[illegible]	Male	Ratnapura	2026-03-28 13:28:15.000	NULL	1

Query executed successfully.   Amaavi\SQLEXPRESS (17.0 RTM)

Step 6: ETL development – Accumulating fact tables

Fact Update Validation

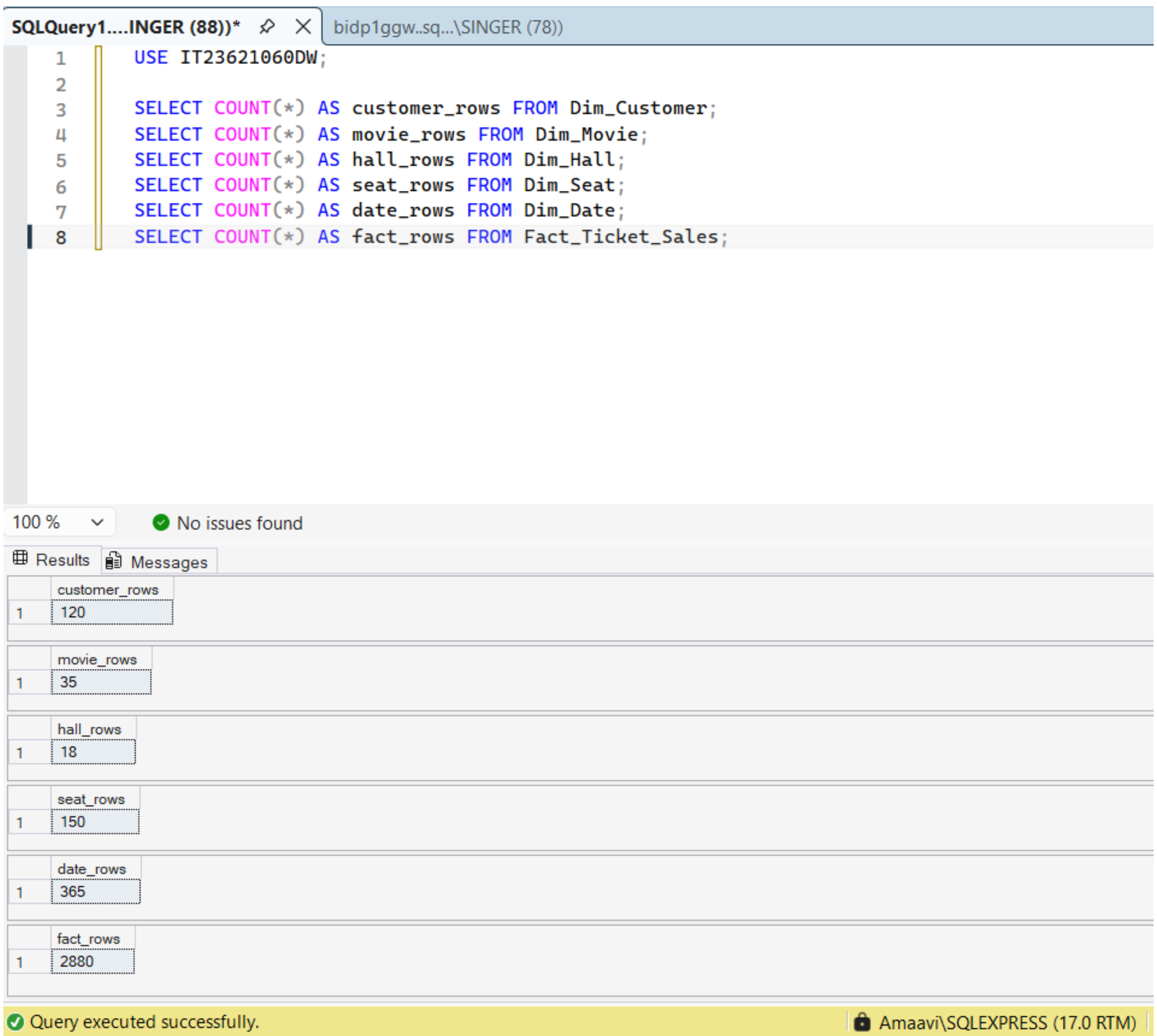
```sql
USE IT23621060DW;

SELECT ticket_id,
       accm_txn_create_time,
       accm_txn_complete_time,
       txn_process_time_hours
FROM Fact_Ticket_Sales
WHERE ticket_id IN ('P0092','H4037','L2502','P4991','J1185',
                    'M6402','S3535','C1073','I7517','Q8198');
```

	ticket_id	accm_txn_create_time	accm_txn_complete_time	txn_process_time_hours
1	P0092	2026-03-25 21:59:07.490	2026-03-27 10:30:00.000	37
2	H4037	2026-03-25 21:59:07.490	2026-03-27 14:00:00.000	41
3	L2502	2026-03-25 21:59:07.490	2026-03-28 09:15:00.000	60
4	P4991	2026-03-25 21:59:07.490	2026-03-28 16:45:00.000	67
5	M6402	2026-03-25 21:59:07.490	2026-03-29 18:20:00.000	93
6	S3535	2026-03-25 21:59:07.490	2026-03-30 08:10:00.000	107
7	C1073	2026-03-25 21:59:07.490	2026-03-30 13:50:00.000	112
8	I7517	2026-03-25 21:59:07.490	2026-03-31 15:00:00.000	138
9	Q8198	2026-03-25 21:59:07.490	2026-03-31 19:40:00.000	142

A separate ETL process was implemented to simulate an accumulating fact table. A CSV file containing ticket completion timestamps was used as input. The OLE DB Command was used to update the fact table by setting the completion time and calculating the transaction processing duration.

Validation

After completing the ETL process, validation was performed to ensure correct data loading.



Conclusion

The data warehouse was successfully designed and implemented using a star schema. The ETL process ensured proper data integration and transformation, while the accumulating fact table demonstrated real-world transaction updates. The solution supports efficient data analysis and business intelligence operations.

References

- https://www.kimballgroup.com/data-warehouse-business-intelligence-resources/kimball-techniques/dimensional-modeling-techniques
- https://learn.microsoft.com/en-us/sql/integration-services/data-flow/transformations/slowly-changing-dimension-transformation?view=sql-server-ver17
- https://learn.microsoft.com/en-us/sql/integration-services/data-flow/transformations/merge-join-transformation?view=sql-server-ver17